



Advances in welding of titanium and its alloys using CMT, MIG, and laser-MIG hybrid processes: A comprehensive review

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ABSTRACT

The need to weld Titanium (Ti) and its alloys is becoming more important in relation to aerospace, medical, and naval industries that require light yet strong structures. The present common fusion process of Ti alloys, however, is limited because it is hampered by issues arising from oxidation, grain coarsening, and porosity. The objective of this paper is to summarize recent findings in relation to Cold Metal Transfer (CMT), Metal Inert Gas (MIG), or Laser-MIG (LAMIG) methodologies used in Ti and its alloys. The CMT and pulse CMT (CMT+P) process supports low heat input with improved microstructures that are achieved by governing droplet transference. The addition of minor oxidative components to gas shielding during welding arc enhancement and improved weld hardness supports improved weld properties. The addition of high-intensity laser processing to groove-assisted weld control in MIG supports improved defect-free welds in relation to strength and ductility. The output parameters include the weld parameters that are directly related to the factors of porosity and plasma stability. The analysis reveals that the hybrid approach is better than conventional methods for enhancing key parameters related to properties. Correlations between process, microstructure, and properties are presented in this work for use in aerospace industries.

1. Introduction

Dissimilar metal joining has become an essential technology across multiple industries where the integration of distinct material properties is critical for performance [1]. In aerospace, this technique enables the development of lightweight yet high-strength structures tailored for demanding service conditions in aircraft and spacecraft [2–5]. In the automotive sector, dissimilar metal joints contribute to reducing overall

weight, enhancing fuel efficiency, and integrating materials with diverse properties to improve vehicle performance [6]. Similarly, in nuclear engineering, joining dissimilar alloys is vital for producing components that withstand extreme environments, facilitate heat dissipation, and optimize neutron absorption [7]. The ability of dissimilar metal joining to provide tailored solutions across such sectors highlights its versatility and significance in advancing both technology and materials science [8].

Abbreviations: AM, Additive Manufacturing; ANSI, American National Standards Institute; AWS, American Welding Society; BCC, Body-Centered Cubic; BM, Base Metal; CMT, Cold Metal Transfer; CMT+P, Cold Metal Transfer + Pulse; CMT-ADV, Advanced Cold Metal Transfer; CMT-PAD, Advanced Pulse Cold Metal Transfer; CP-Ti, Commercially Pure Titanium; CO₂, Carbon Dioxide; DWJ, Dual-Phase Weld Joint; EBW, Electron Beam Welding; ELI, Extra Low Interstitial; FLUENT, Computational Fluid Dynamics Software; GMAW, Gas Metal Arc Welding; HAZ, Heat-Affected Zone; HCP, Hexagonal Close-Packed; ICME, Integrated Computational Materials Engineering; JIS, Japanese Industrial Standards; L-DED, Laser Directed Energy Deposition; LAMIG, Laser-Metal Inert Gas Hybrid Welding; LBW, Laser Beam Welding; MIG, Metal Inert Gas Welding; ML, Machine Learning; NG-LAHW, Narrow Gap Laser-Arc Hybrid Welding; O₂, Oxygen; PBF, Powder Bed Fusion; P-MIG, Pulsed Metal Inert Gas Welding; PMC, Pulse Multi-Control; SAW, Surface Acoustic Wave; SEM, Scanning Electron Microscopy; Ti, Titanium; Ti64, Ti-6Al-4V Alloy; TIG, Tungsten Inert Gas Welding; TMP, Thermomechanical Processing; UTS, Ultimate Tensile Strength; UIT, Ultrasonic Impact Treatment; VOF, Volume of Fluid Model; WAAM, Wire Arc Additive Manufacturing; WES, Welding Engineering Standard.

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Among advanced structural materials, titanium and its alloys have gained particular prominence. Their widespread use can be attributed to significant progress in titanium metallurgy and the concurrent development of reliable welding methods. Titanium alloys are generally weldable by several fusion welding techniques, with inert gas-shielded arc welding and high-energy beam processes being the most widely employed [9–11]. Metal inert gas (MIG) welding remains the predominant technique, followed by tungsten inert gas (TIG) welding. However, despite its popularity, the TIG process is not always practical for large-scale components due to cost and operational constraints [12–14]. MIG welding, although attractive for productivity, has been limited in application for titanium due to issues such as poor arc stability and excessive spatter [15].

The main challenges in arc welding titanium alloys arise from their high reactivity at elevated temperatures. In liquid states, titanium readily absorbs oxygen, nitrogen, and hydrogen, which can induce embrittlement of the weld. Furthermore, excessive grain growth at high temperatures can significantly compromise both toughness and corrosion resistance of the weld metal [16–19]. To address these issues, advanced welding technologies such as Cold Metal Transfer (CMT) [20–23], electron beam welding (EBW) [24–26], and hybrid laser beam welding (LBW) [27–31] have been investigated. These processes offer several advantages over conventional fusion welding, including low heat input, high joint reproducibility, minimal distortion, and the ability to produce full-penetration welds in a single pass. In particular, laser-MIG and laser-TIG hybrid processes have shown promise for welding titanium alloys, eliminating the need for costly vacuum chambers [32,33].

Despite the significant advantages offered by advanced welding techniques such as Cold Metal Transfer (CMT), Electron Beam Welding (EBW), and Laser Beam Welding (LBW), several inherent limitations restrict their widespread industrial implementation. High capital investment and operational costs remain a major barrier, particularly for EBW and LBW systems, which require sophisticated equipment such as high-vacuum chambers or high-power laser sources. Additionally, EBW is constrained by vacuum requirements, limiting its applicability for large-scale or in-situ structures. Similarly, LBW and laser-hybrid processes demand precise alignment, strict process control, and skilled operation, which increase system complexity and reduce process robustness in industrial environments [34,35]. Although CMT offers lower heat input and improved weld quality, its deposition rate and productivity may be lower compared to conventional high-energy processes in certain applications. These challenges highlight the trade-off between process precision and industrial scalability, necessitating further development of cost-effective and adaptable welding solutions [34,35].

A major breakthrough in this context was the development of Cold Metal Transfer (CMT) welding by Fronius in 2004 [36,37]. CMT represents an advanced variant of the MIG/MAG process [38], distinguished by its low heat input approximately 10 % of that in conventional MIG welding [39]. This reduction in thermal load minimizes burn-through, reduces distortion, and refines the weld microstructure through a controlled stepwise droplet transfer mechanism [40]. Despite these advantages, the post-weld microstructure often remains heterogeneous, necessitating precise heat treatments to stabilize dual-phase weld joints (DWJ). Effective control of post-weld thermal cycles is crucial for optimizing microstructure stability, toughness, and corrosion resistance in ferritic / martensitic-austenitic systems exposed to elevated temperatures and pressures [41].

Technically, the CMT process employs a three-phase low-temperature cycle, wherein molten droplet transfer is achieved through controlled wire retraction synchronized with the arc current [42]. This results in significantly reduced spatter, lower dilution (3–4 %), and high material utilization efficiency, making it particularly suitable for hard-facing applications such as Stellite 21 on H13 steel [43]. Comparative studies further demonstrate that, at an equivalent filler wire feed rate,

CMT achieves lower dilution in aluminum alloys compared with conventional gas metal arc welding (GMAW) [44]. Owing to its advantages of high joint strength, superior microstructure, and capability to weld dissimilar materials, CMT is increasingly adopted in aerospace, automotive, bridge construction, microelectronics, and railway industries [45,46]. Moreover, the introduction of advanced variants such as pulse-CMT (CMT-P), advanced CMT (CMT-ADV), and advanced pulse CMT (CMT-PAD) further extend its application potential [47].

The mechanism of CMT welding relies on high-frequency pulsing to transfer molten droplets into the weld zone, where the arc is briefly quenched to minimize heat input [48]. The process typically employs reverse polarity and helium-based shielding gas mixtures, which together stabilize the arc and further reduce spatter formation [49–51]. The CMT cycle can be divided into three phases: peak current, background current, and voltage short-circuiting [52–54]. During short-circuiting, the filler wire is retracted by a microprocessor, breaking the circuit and significantly lowering thermal input. This reduced heat promotes refinement of the intermetallic layer and improves overall mechanical properties. Critical parameters such as pulse frequency and wire retraction speed directly influence grain size and tensile strength of the weld [55–59].

Nevertheless, the underlying physics of CMT remain complex. According to Mezrag et al. [46], the intricate waveform and filler back-feeding introduce challenges in fully understanding the coupling between current, heat transfer, and metal transfer. Further studies are required to elucidate these relationships. Despite these challenges, the process demonstrates superior arc length control, high edge coupling tolerance, and the ability to join thin sheets with minimal distortion, allowing it to outperform conventional methods in many industrial applications [60].

By contrast, gas metal arc welding (GMAW) the broader category encompassing MIG remains one of the most widely applied welding processes worldwide. Initially developed for high deposition rates and productivity, GMAW offers ease of automation and integration with robotics, making it suitable for large-scale production [61,62]. It is highly versatile, accommodating a broad range of materials and thicknesses [39]. However, limitations such as porosity, fusion defects, and high fume generation continue to restrict its use for critical titanium applications. Additionally, its limited control over metal transfer and higher heat input compared to CMT underscore the importance of adopting advanced processes [22,35,62,63].

In addition to similar-alloy joining, dissimilar welding of titanium alloys has emerged as a major research focus because modern aerospace, chemical, marine, and biomedical structures increasingly require titanium to be joined with lightweight or cost-effective dissimilar materials such as aluminum alloys and steels. However, each dissimilar combination presents distinct metallurgical and process-related challenges. For example, Ti/Al joints are highly prone to the formation of brittle Ti-Al intermetallic compounds, while Ti/steel joints often suffer from severe metallurgical incompatibility, thermal expansion mismatch, and formation of brittle Fe-Ti reaction products [64–66]. These challenges necessitate process-specific solutions involving low heat input, controlled dilution, optimized filler or interlayer materials, and precise thermal management. In this context, advanced processes such as CMT, pulsed MIG, and Laser-MIG hybrid welding are increasingly attractive because they offer better control over heat input, fusion behavior, droplet transfer, and interfacial reactions than conventional arc welding routes. Accordingly, understanding the relationship between dissimilar material combination, reaction mechanism, and process selection is essential for the development of reliable titanium-based multi-material joints [67–69].

The growing industrial demand for titanium alloys further emphasizes the importance of advanced joining methods. Titanium alloys are widely used in aerospace, chemical, marine, and biomedical applications due to their high strength-to-weight ratio, excellent corrosion resistance, low density, good formability at elevated temperatures, and

exceptional biocompatibility [70–73]. Their increasing adoption in aerospace has been driven by the replacement of older, less fuel-efficient aircraft. In such applications, materials must endure complex combinations of static and dynamic stresses, high thermal loads, and aggressive service environments [5,74–77]. In addition to their intrinsic properties, titanium alloys offer distinct comparative advantages over conventional structural materials such as steels, aluminum alloys, and nickel-based superalloys. Compared with steels, titanium alloys provide comparable strength at nearly half the density, resulting in superior specific strength, which is critical for aerospace weight reduction. Unlike aluminum alloys, titanium exhibits significantly higher corrosion resistance, particularly in chloride-rich and marine environments, as well as improved performance at elevated temperatures beyond 300 °C. Furthermore, in comparison to nickel-based alloys, titanium alloys offer lower density and better biocompatibility, making them more suitable for biomedical and lightweight structural applications. These performance criteria specific strength, corrosion resistance, thermal stability, and biological compatibility collectively justify the widespread adoption of titanium alloys in high-performance engineering sectors.

Among these, Ti-6Al-4V (Ti64) dominates, accounting for 80–90 % of titanium usage in aerospace structures [77,78]. As an (α + β) alloy, Ti64 combines high fracture toughness, excellent corrosion resistance, and superior biocompatibility, making it the material of choice for components ranging from fuselages, skin panels, and landing gears to turbine fan blades and engine disks [77–81]. Commercially pure (CP) titanium also retains significance, particularly Grade 2, which balances strength, corrosion resistance, and ductility. While CP titanium offers outstanding biocompatibility and corrosion resistance, its relatively low strength restricts applications in high-load biomedical implants. In contrast, Ti-6Al-4V is widely used in biomedical devices, though concerns persist over the potential toxicity of alloying elements such as aluminum and vanadium during long-term implantation [82–84]. While Ti-6Al-4V remains the most extensively studied titanium alloy, the influence of alloy chemistry and impurity content on weld behavior is equally critical. Interstitial elements such as oxygen, nitrogen, and hydrogen significantly affect weld integrity by increasing hardness and promoting embrittlement, particularly in the fusion and heat-affected zones. Similarly, variations in alloying elements (e.g., Al as an α -stabilizer and V as a β -stabilizer) alter the β -transus temperature, thereby influencing phase transformation kinetics and microstructural evolution during welding. These compositional effects govern the formation of α' martensite, retained β -phase, and grain morphology, ultimately determining mechanical performance and residual stress distribution.

To overcome these challenges, researchers have sought to enhance CP titanium through thermo-mechanical processing, including rolling, extrusion, and wire drawing, which refine grain size, improve texture, and optimize mechanical properties [85]. Understanding the evolution of microstructure and deformation mechanisms in CP titanium thus remains a critical research area, not only for biomedical applications but also for improving its suitability in demanding aerospace and marine environments.

Additionally, the initial fabrication route of titanium alloys (e.g., forging, casting, or additive manufacturing) plays a critical role in defining the microstructural state and weldability, necessitating integrated consideration of processing and joining strategies.

In recent years, significant progress has been made in the development of advanced welding technologies for titanium and its alloys, driven largely by the increasing demand for lightweight and high-performance materials in aerospace, biomedical, and energy applications. Contemporary research has focused on improving weld quality through better control of heat input, droplet transfer behavior, and melt pool dynamics [86]. Among modern arc welding techniques, Cold Metal Transfer (CMT) has emerged as a promising low-heat-input process capable of producing stable metal transfer and refined weld microstructures, particularly in Ti-6Al-4V alloys. Several recent investigations have reported that advanced variants such as pulsed-CMT and CMT-ADV

can significantly reduce spatter formation, minimize distortion, and improve mechanical properties by promoting finer α' martensitic structures in the fusion zone [34,87–89].

In parallel, continued developments in pulsed Metal Inert Gas (MIG) welding have improved arc stability and droplet transfer control, enabling better weld penetration and reduced defect formation in titanium alloys. Researchers have also explored optimized shielding gas compositions and parameter control strategies to mitigate contamination and porosity during welding. Furthermore, hybrid welding technologies, especially Laser-MIG hybrid welding, have gained considerable attention due to their ability to combine the deep penetration capability of laser beams with the flexibility of arc welding filler deposition. Recent experimental and numerical studies have demonstrated that such hybrid processes can significantly improve weld morphology, reduce porosity formation, and achieve a favorable strength-ductility balance in titanium alloy joints [27,90,91].

These recent developments highlight the rapid evolution of titanium welding technologies and emphasize the importance of understanding the metallurgical characteristics of titanium alloys, including phase transformations, microstructural stability, and alloy classification. A comprehensive understanding of these metallurgical aspects is therefore essential for interpreting welding behavior and optimizing process parameters. Accordingly, the following section discusses the fundamental physical metallurgy and classification of titanium and its alloys, which form the basis for understanding their welding characteristics [92].

2. Metallurgical aspects of titanium and its alloys

2.1. Physical metallurgy

Titanium (Ti) exhibits two distinct allotropic modifications depending on temperature: the hexagonal close-packed (HCP) α -phase, which is stable at lower temperatures, and the body-centered cubic (BCC) β -phase, which exists at elevated temperatures. The transformation between these phases occurs at approximately 882 °C, a critical temperature commonly referred to as the β -transus. At this point, the crystal structure undergoes a transition from HCP (α) to BCC (β), with the latter remaining stable up to the melting point of titanium [79, 93–95]. The lattice parameters and crystal structures of the α and β -phases are depicted in Fig. 1. Notably, titanium exhibits a c/a ratio of 1.587 in its α -phase, which is slightly lower than the ideal value of 1.632 for an HCP crystal structure [93,96–98].

The β -transus temperature of titanium is highly sensitive to alloying additions, as illustrated in Fig. 2. Alloying elements are typically classified based on their effect on phase stability. Elements such as aluminum (Al), oxygen (O), carbon (C), and nitrogen (N) act as α -stabilizers by increasing the β -transus temperature and thereby extending the stability range of the α -phase. In contrast, β -stabilizers decrease the β -transus temperature by promoting the stability of the β -phase [99]. β -stabilizers are further subdivided into two categories: β -isomorphous and β -eutectoid. Elements such as molybdenum (Mo), niobium (Nb), vanadium (V), and tantalum (Ta), which exhibit high solubility in titanium, belong to the β -isomorphous category. Conversely, elements with limited solubility, including iron (Fe), manganese (Mn), chromium (Cr), cobalt (Co), nickel (Ni), and hydrogen (H), form intermetallic compounds with titanium and are classified as β -eutectoid stabilizers [100–102]. Additionally, neutral elements such as tin (Sn) and zirconium (Zr) exert negligible influence on the β -transus temperature; however, they are widely utilized for solid-solution strengthening of titanium alloys [93–95].

2.2. Classification of titanium alloys

Titanium alloys are broadly categorized into three groups α , $\alpha + \beta$, and β alloys based on the predominant phases present at room temperature, as illustrated in Fig. 3 [103]. Within these categories, further

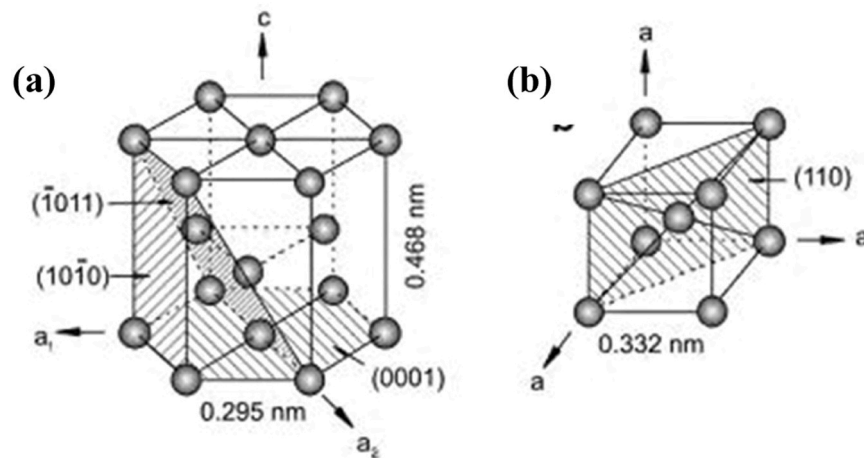


Fig. 1. HCP α -phase and BCC β -phase crystal structures of titanium (a) Hexagonal close packed (α phase), (b) Body centered cubic (β phase) [79,94].

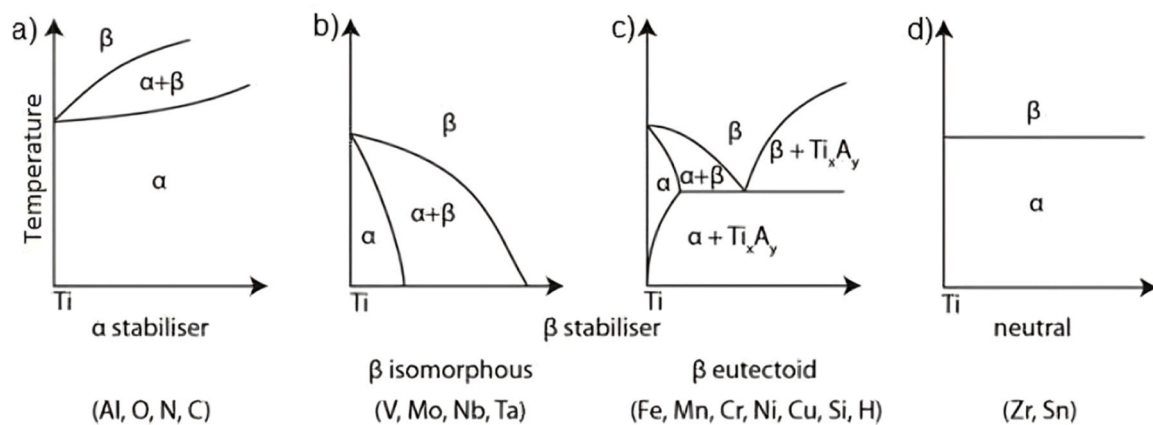


Fig. 2. Effect of alloying elements on phase diagrams of titanium alloys (a) α -stabilizers, (b) β -isomorphous stabilizers, (c) β -eutectoid stabilizers and (d) neutral stabilizers [95].

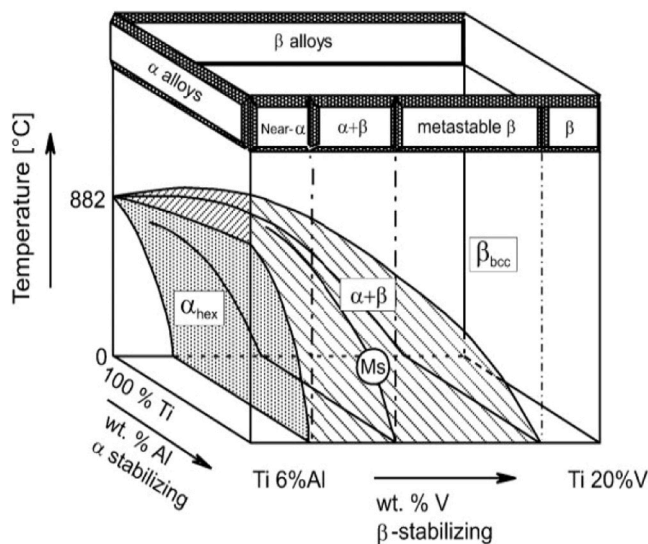


Fig. 3. Classification of α , $\alpha+\beta$, and β titanium alloys [103].

sub-classifications exist, namely near- α alloys and metastable β alloys. The α and near- α titanium alloys are primarily developed by incorporating α -stabilizing elements. These alloys are well recognized for their

superior corrosion resistance and excellent formability; however, their relatively limited strength constrains their application mainly to the chemical, biomedical, and marine industries [104,105]. In contrast, β and metastable β alloys are stabilized through the addition of sufficient β -stabilizing elements to retain a fully β structure and suppress martensitic transformation upon quenching from the β phase field [94, 106,107]. These alloys exhibit high strength, toughness, and superior fatigue resistance compared with other titanium alloy systems, while also being amenable to heat treatment [78,95]. Nevertheless, their wider adoption in industry is restricted due to drawbacks such as relatively high density, low modulus, and elevated production costs. These alloys are well recognized for their superior corrosion resistance (often exhibiting corrosion rates <0.01 mm/year in aggressive environments) and excellent formability; however, their relatively limited strength (~ 300 – 600 MPa) constrains their application mainly to chemical, biomedical, and marine industries [108].

In addition to qualitative descriptions, the mechanical properties of different titanium alloy classes exhibit significant quantitative variations. Commercially pure titanium (CP-Ti, Grade 2) typically shows a yield strength of ~ 275 – 450 MPa and ultimate tensile strength (UTS) of ~ 345 – 550 MPa, with elongation exceeding 20 %, indicating excellent ductility and corrosion resistance but limited load-bearing capability [109,110]. In contrast, $\alpha+\beta$ alloys such as Ti-6Al-4V exhibit significantly higher strength, with yield strength ~ 880 MPa and UTS ~ 950 – 1100 MPa, along with moderate elongation (~ 10 – 15 %). β -type titanium alloys (e.g., Ti-10V-2Fe-3Al) demonstrate even higher strength

levels, with UTS exceeding 1100–1300 MPa, while maintaining good fracture toughness and fatigue resistance, albeit with relatively lower corrosion resistance compared to CP-Ti [111]. These quantitative differences highlight the trade-off between strength, ductility, and corrosion resistance across titanium alloy systems, which directly influences their weldability and application domains.

$\alpha + \beta$ titanium alloys represent the most widely used group among commercial titanium alloys due to their exceptional combination of high strength, good corrosion resistance, and moderate ductility. At room temperature, these alloys typically contain a β -phase volume fraction ranging from 5 % to 40 % [74,78,79,112–114]. Among them, Ti-6Al-4V stands out as the most important $\alpha + \beta$ alloy, offering a unique balance of properties such as high specific strength, excellent fatigue and fracture resistance at elevated temperatures, and superior corrosion resistance. Owing to this versatile performance, more than 50 % of titanium alloys used commercially fall within the Ti-6Al-4V family, which finds applications across aerospace, biomedical, chemical, and other engineering sectors. In addition to Ti-6Al-4V, other $\alpha + \beta$ alloys such as Ti-6Al-2Sn-4Zr-2Mo, Ti-550, and Ti-662 are employed in high-strength and elevated-temperature applications [94,115]. The key physical and mechanical properties of Ti-6Al-4V are summarized in Tables 1, 2 and Table 3 [94,115]. The microstructural characteristics of $\alpha + \beta$ titanium alloys are crucial in determining their mechanical performance. In Ti-6Al-4V, three distinct microstructural types are commonly observed: lamellar, equiaxed, and duplex (Fig. 4) [79].

- **Lamellar microstructure:** This structure is obtained by heat treatment within the high-temperature β -phase field followed by slow cooling into the $\alpha + \beta$ phase field. The strength of lamellar structures is governed by the thickness of α -lamellae, which is strongly dependent on the cooling rate. Finer lamellae contribute to higher strength, whereas coarser lamellae enhance resistance to fatigue crack propagation, thereby improving fracture toughness [118–120].
- **Equiaxed microstructure:** Achieved through extensive thermo-mechanical processing within the $\alpha + \beta$ temperature range, equiaxed structures are characterized by the breakdown of α -lamellae into equiaxed grains. Grain refinement to sizes as small as $\sim 2 \mu\text{m}$ can be achieved, resulting in superior strength compared with the lamellar and duplex counterparts [121–124].
- **Duplex microstructure:** Also referred to as bimodal, the duplex structure is typically formed by quenching from the $\alpha + \beta$ phase field (approximately 750–1000 °C). During quenching, the β -phase transforms into α' or α'' martensite, while the primary α -phase remains unchanged. The overall strength of the duplex microstructure depends on both the lamellar size and the volume fraction of the retained primary α -phase. Generally, duplex structures provide a higher strength level compared to lamellar microstructures, offering a balance between strength and toughness [125–128].

In addition to their morphological characteristics, the lamellar, equiaxed, and duplex microstructures exhibit distinctly different mechanical responses due to variations in phase distribution, grain size, and α/β interface characteristics. The lamellar microstructure,

Table 1
Physical properties of Ti6Al4V alloy [94,115].

Physical Properties	Values
Density	4.43 g/cc
Melting point	1650 ± 15 °C
Beta transus	995 ± 15 °C
Thermal conductivity (at room temp.)	6.6 W/m K
Thermal conductivity (at 400 °C)	13 W/m K
Specific heat (at room temp.)	0.57 J/g K
Specific heat (at 400 °C)	0.65 J/g K
Thermal expansion (0–300 °C)	9.5×10^{-6} K ⁻¹

Table 2
Mechanical properties of Ti6Al4V alloy [94,115–117].

Mechanical Properties	Values	Mechanical Properties	Values
Yield strength	880 MPa	Ultimate shear strength	550 MPa
Ultimate tensile strength	950 MPa	Shear modulus	44 GPa
Elongation at break	14 %	Hardness (Vickers)	349 HV
Reduction of area	36 %	Poisson's ratio	0.3–0.33
Modulus of Elasticity	114 GPa		

Table 3
Comparative Mechanical Properties of Titanium Alloys.

Alloy Type	Yield Strength (MPa)	UTS (MPa)	Elongation (%)	Key Feature
CP-Ti	275–450	345–550	20–30	Excellent corrosion resistance
Ti–6Al–4V	~ 880	950–1100	10–15	Balanced strength & ductility
β -Ti alloys	900–1200	1100–1300	8–12	High strength & fatigue resistance

consisting of coarse α -lamellae within prior β grains, typically demonstrates superior fracture toughness and fatigue crack growth resistance, owing to the tortuous crack propagation path and effective crack deflection at α/β interfaces. However, its relatively coarse structure results in lower yield strength and ductility compared to refined microstructures [129].

In contrast, the equiaxed microstructure, characterized by fine and uniformly distributed α grains, provides enhanced yield strength and ductility due to grain boundary strengthening and homogeneous deformation behavior. This microstructure is particularly beneficial for applications requiring improved tensile properties and isotropic mechanical performance.

The duplex (bimodal) microstructure offers a balanced combination of strength, ductility, and fracture toughness by integrating equiaxed primary α grains within a transformed β matrix containing fine lamellar α . This hybrid structure effectively combines the advantages of both lamellar and equiaxed morphologies, making it highly suitable for aerospace applications where both strength and damage tolerance are critical [130,131].

These microstructures are strongly dependent on processing routes. Lamellar structures are typically obtained through β -annealing followed by slow cooling, equiaxed microstructures are produced via thermo-mechanical processing within the $\alpha + \beta$ phase field, and duplex structures are achieved through controlled heat treatment and quenching from the $\alpha + \beta$ region. In welding processes such as CMT, MIG, and Laser-MIG hybrid welding, the thermal cycle particularly peak temperature and cooling rate plays a decisive role in determining the resulting microstructure. Rapid cooling during welding often promotes α' martensitic transformation, while controlled heat input and post-weld heat treatment can tailor the microstructure toward lamellar or duplex configurations, thereby influencing the final mechanical performance of the joint [132,133]. Table 4 summarized in Comparative Microstructure Properties of Titanium Alloys.

In addition to primary alloying elements, the presence of trace impurities plays a decisive role in weld quality. Oxygen and nitrogen act as strong α -stabilizers and increase strength at the expense of ductility, while hydrogen may lead to porosity and delayed cracking. During welding, the high-temperature exposure enhances the diffusion and absorption of these elements, making shielding conditions and material purity crucial. Variations in alloy composition can also shift the β -transus temperature, altering cooling rates and phase transformation pathways, which directly impacts weld microstructure and performance.

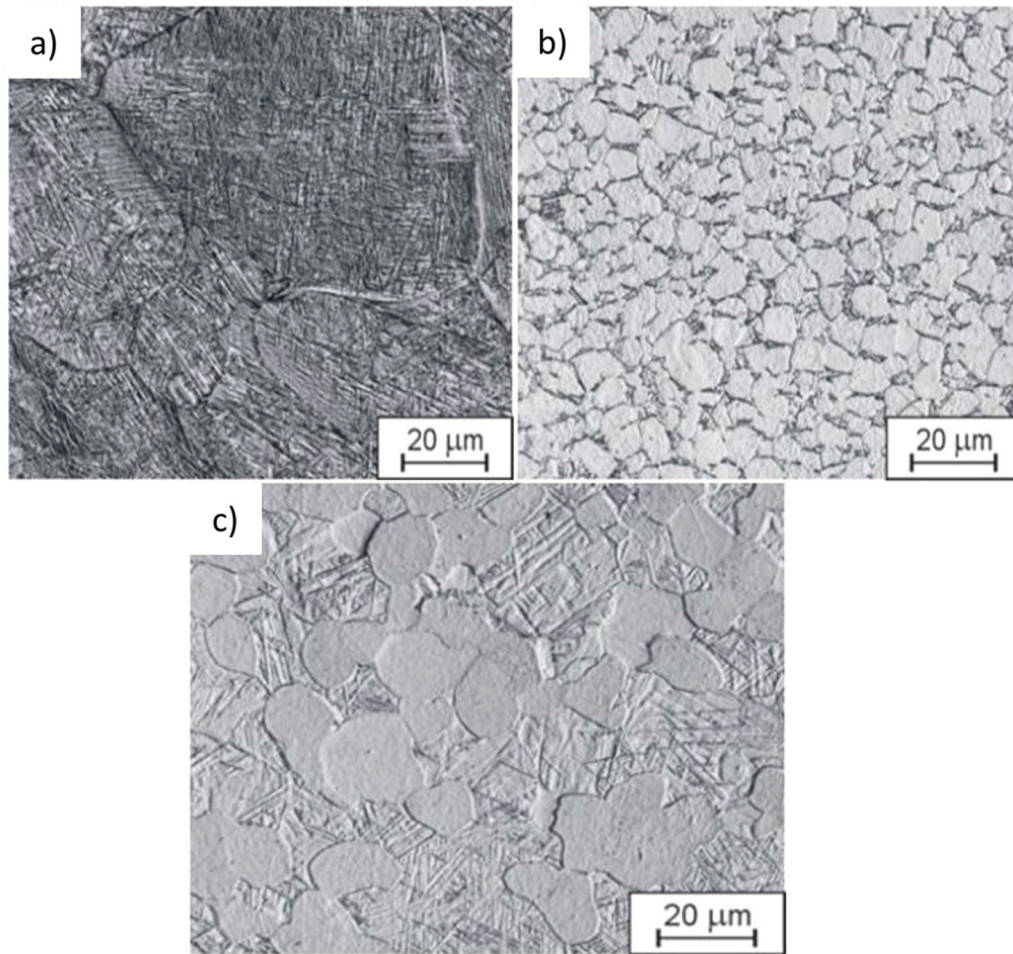


Fig. 4. Typical microstructures (lamellar, equiaxed, and bimodal microstructures) observed in Ti6Al4V alloy: (a) Lamellar, (b) Equiaxed and (c) Bimodal [134].

Table 4
Comparative Microstructure Properties of Titanium Alloys.

Microstructure	Processing Route	Strength	Ductility	Fatigue Resistance	Toughness
Lamellar	β -annealing + slow cooling	Moderate	Low	High	High
Equiaxed	TMP in α + β region	High	High	Moderate	Moderate
Duplex	α + β heat treatment + quenching	High	Moderate-High	High	High

2.3. Applications of Ti6Al4V alloy

The selection of Ti-6Al-4V over other engineering materials is primarily governed by performance-driven criteria. Its high strength-to-weight ratio surpasses that of most steels and aluminum alloys, enabling structural efficiency in aerospace systems. Its excellent resistance to corrosion and oxidation makes it superior to steels in marine and chemical environments. Additionally, its ability to retain mechanical properties at moderately elevated temperatures provides an advantage over aluminum alloys, while its superior biocompatibility distinguishes it from nickel-based alloys in medical applications. These comparative advantages make Ti-6Al-4V a material of choice where performance, durability, and weight reduction are simultaneously required [135].

Ti-6Al-4V (Ti64) alloy has emerged as a material of paramount importance in the aerospace sector owing to its exceptional combination of properties, including a high strength-to-weight ratio, superior fatigue resistance, and excellent corrosion resistance. One of the most attractive features of Ti64 is its ability to retain mechanical strength at elevated temperatures, typically up to 300–400 °C, making it highly suitable for

service in demanding environments [78,80,136,137]. Consequently, this alloy is extensively employed in manufacturing low-pressure compressor fan blades and discs for aviation engines, as well as critical aircraft structural components such as fuselage sections, wings, and landing gear. Forged Ti64 is also a preferred material for rotor heads in helicopters, where both strength and weight reduction are crucial [9, 138,139].

Beyond aerospace applications, Ti64 finds widespread use in diverse industries due to its unique mechanical and chemical stability. For instance, in the chemical, oil, and gas sectors, it is applied in mining tools, drilling equipment, and risers, particularly for offshore petroleum operations, where resistance to corrosion and fatigue is vital [9,140]. Its high specific strength and excellent ballistic performance have also attracted considerable attention from the defense industry, particularly for use in military vehicle armor [79,80]. The alloy's outstanding biocompatibility and strength make it a reliable material for biomedical applications, particularly in orthopedic and dental implants, where it is utilized for bone replacement and prosthetics. Furthermore, in energy systems such as power plants, Ti64 is used for manufacturing steam turbine components that must endure high stress and harsh operating

environments [141,142]. Its excellent corrosion resistance and low elastic modulus also enable its application in ship pressure hulls for deep-sea oil platforms [143]. In addition to industrial and medical uses, Ti64 has also penetrated the consumer and sports markets [144]. Its combination of low density, strength, and durability has facilitated its use in high-performance sporting goods, including hollow golf club heads, racing bicycles, and mountain bikes [145,146]. Overall, the versatility of Ti-6Al-4V across aerospace, defense, biomedical, energy, and recreational applications underscores its position as one of the most valuable structural alloys available today [78,80]. Various examples of Ti6Al alloy is given below in Fig. 5.

In recent years, significant research efforts have been directed toward improving welding technologies for titanium alloys through advanced arc control, hybrid energy sources, and auxiliary process modifications. Modern techniques such as Cold Metal Transfer (CMT), pulsed MIG welding, and Laser-MIG hybrid welding have been widely investigated due to their capability to reduce heat input, control droplet transfer behavior, and improve weld microstructure. Recent studies have also explored the use of ultrasonic vibration, magnetic fields, and optimized shielding gas compositions to enhance arc stability, reduce porosity formation, and improve the mechanical performance of titanium alloy joints. These developments highlight the rapid evolution of welding technologies and the importance of understanding the process microstructure property relationships in advanced titanium welding systems. Table 5 summarized Comparative Performance of Titanium Alloys vs Conventional Materials [147–149].

2.4. Fabrication methods of titanium alloys and their relevance to welding

Titanium alloys are processed through a variety of fabrication routes, including casting, forging, thermomechanical processing, and additive manufacturing, each of which significantly influences the resulting microstructure, mechanical properties, and weldability of the material [150].

Casting is one of the primary fabrication methods used for titanium components, particularly for complex geometries in aerospace and

Table 5
Comparative Performance of Titanium Alloys vs Conventional Materials.

Property	Titanium Alloys (Ti–6Al–4V)	Steel	Aluminum Alloys	Ni-based Alloys
Density (g/cm ³)	~4.43	~7.8	~2.7	~8.2
Specific Strength	Very High	Moderate	High	Moderate
Corrosion Resistance	Excellent	Moderate-Poor	Moderate	Excellent
High Temperature Performance	Up to 400°C	Moderate	Poor	Excellent (>700°C)
Biocompatibility	Excellent	Poor	Moderate	Poor
Fatigue Resistance	Excellent	Good	Moderate	Good

biomedical applications. However, due to titanium’s high reactivity at elevated temperatures, casting often leads to issues such as contamination, porosity, and coarse prior-β grain structures. These characteristics can adversely affect weld quality by promoting defect formation and heterogeneous microstructures in the fusion zone [151,152].

Forging and thermomechanical processing (e.g., rolling, extrusion) are widely employed to refine grain structure and improve mechanical properties. Forged Ti-6Al-4V typically exhibits equiaxed or bimodal microstructures with reduced defect density, which enhances weldability by providing uniform heat distribution and minimizing susceptibility to cracking. Furthermore, controlled thermomechanical treatments enable tailoring of α/β phase balance, directly influencing phase transformations during welding [116].

In recent years, additive manufacturing (AM) techniques such as Wire Arc Additive Manufacturing (WAAM), Laser Directed Energy Deposition (L-DED), and Powder Bed Fusion (PBF) have gained significant attention for titanium alloys. These processes produce unique microstructures characterized by columnar prior-β grains and anisotropic mechanical properties due to directional solidification. In particular, WAAM using Cold Metal Transfer (CMT) offers lower heat input and reduced residual stresses compared to conventional arc-based deposition methods [153,154].

The fabrication route strongly dictates the initial microstructural

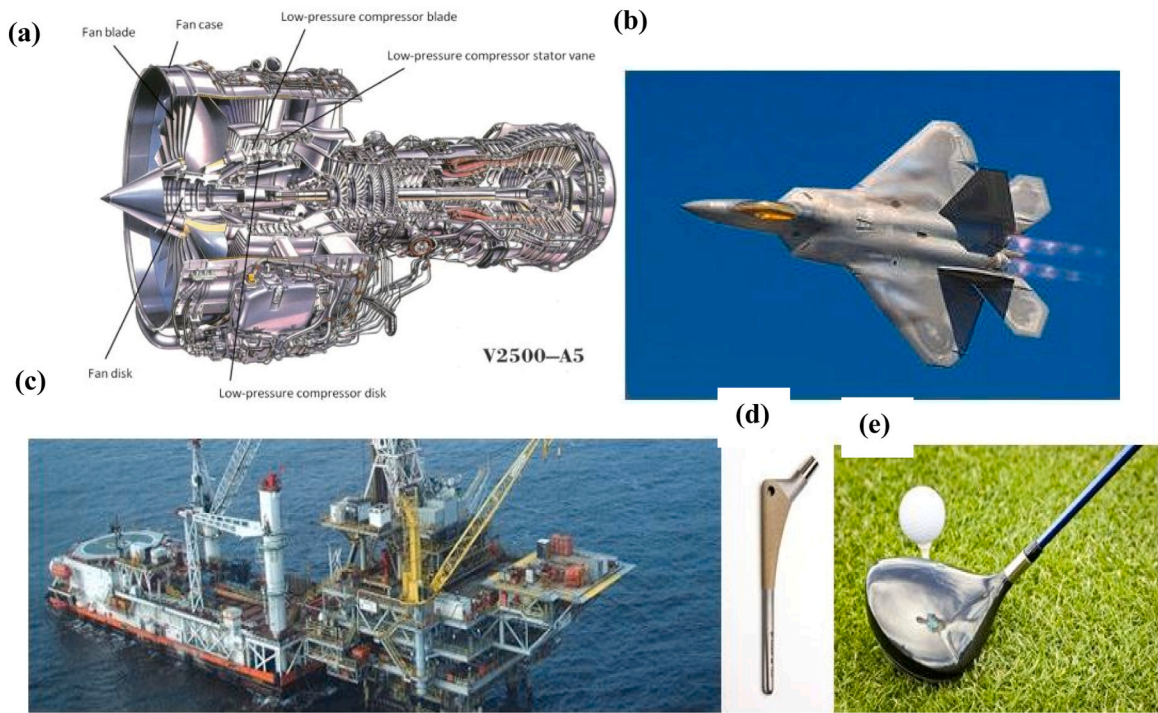


Fig. 5. Examples for various applications of Ti6Al4V alloy. (a) V2500 engine of A320 airbus, (b) Outer body of military aircraft, (c) Deep sea platforms oil and gas mining (d) Hip implant and (e) Golf club heads [79,80].

state, which in turn governs thermal response during welding. For instance, additively manufactured titanium alloys often exhibit higher residual stresses and anisotropy, requiring careful control of welding parameters to avoid distortion and cracking. Similarly, cast structures with coarse grains may lead to uneven heat distribution and increased porosity during welding [155].

Therefore, understanding the interplay between fabrication methods and welding processes is essential for optimizing joint performance. Integrating fabrication-aware welding strategies can significantly enhance microstructural stability, reduce defect formation, and improve mechanical integrity in titanium alloy components used in critical applications.

2.5. Emerging Trends and Innovations in Titanium Alloy Development

Recent advancements in titanium alloy development are increasingly driven by the demand for high-performance materials in aerospace, biomedical, and energy sectors. One of the most significant trends is the development of metastable β titanium alloys, which exhibit superior strength, improved formability, and enhanced fatigue resistance compared to conventional $\alpha+\beta$ alloys. Alloying elements such as Nb, Mo, Ta, and Fe are being strategically incorporated to tailor phase stability and mechanical performance [156–158].

Another important direction involves the design of biocompatible titanium alloys by replacing potentially toxic elements such as Al and V in Ti-6Al-4V with non-toxic alternatives (e.g., Ti-Nb, Ti-Zr-Ta systems). These alloys demonstrate improved corrosion resistance and reduced ion release, making them highly suitable for long-term biomedical implants [159,160].

In parallel, additive manufacturing (AM) has emerged as a transformative approach for titanium processing. Techniques such as WAAM and laser-based deposition are enabling the fabrication of complex geometries with tailored microstructures. This has led to the development of AM-specific titanium alloys, optimized for controlled solidification, reduced residual stress, and improved interlayer bonding [161].

Recent research also emphasizes the role of Integrated Computational Materials Engineering (ICME) and machine learning (ML) in accelerating alloy design. These approaches enable predictive modeling of phase evolution, mechanical properties, and weldability, thereby reducing experimental costs and development time [162].

Furthermore, the development of high-temperature titanium alloys for next-generation aerospace engines is gaining attention. Alloy systems reinforced with silicides or intermetallic phases are being explored to enhance creep resistance and oxidation stability at temperatures exceeding 600 °C.

Despite these advancements, several challenges remain, including maintaining microstructural stability during welding, controlling residual stresses, and minimizing defect formation in advanced alloys. Therefore, the integration of alloy design with advanced welding processes such as CMT and Laser-MIG hybrid welding is essential for achieving optimized performance.

3. Different similar and dissimilar welding process

3.1. CMT + P welding

Guancheng Zhao. [163] study has focused on improving the welding process of Ti-6Al-4V alloy by using ultrasonic vibration in cold metal transfer pulse (CMT + P) welding. This new approach aims to enhance grain structures and strengthen the tensile properties of weld joints. Experimental findings show that ultrasonic vibration effectively prevents columnar crystal growth in the weld zone, leading to improved grain structures and ultimately enhancing the ultimate tensile strength (UTS) shown in Fig. 6. This method significantly increases ultimate tensile strength by refining grain size, resulting in a combination of ductile and brittle fracture modes within the weld zone.

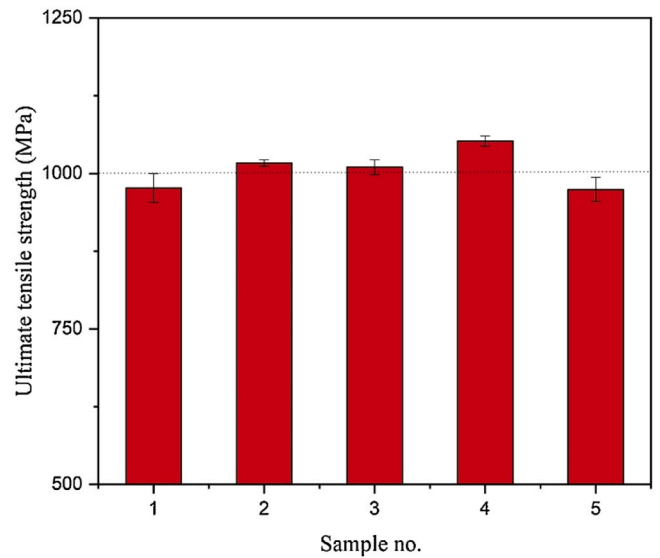


Fig. 6. Tensile tests of Ti-6Al-4V joints prepared via CMT + P welding [163].

The tungsten needle used for ultrasonic vibration in the molten pool was carefully controlled to maintain process stability and optimize welding parameters such as speed and feed. Microstructural and mechanical analyses confirm that this technique positively influences grain growth orientation and mechanical performance without affecting the chemical composition of the weld zone. Recent investigations into ultrasonic-assisted CMT + P welding of Ti-6Al-4V alloy reveal that positioning the ultrasonic-vibrating tungsten needle closer to the rear of the molten pool enhances grain refinement by inhibiting columnar grain growth. The microstructure of the weld zone primarily consists of α' martensite, α phase, and residual β phase, with ultrasonic vibration influencing grain orientation and microstructural phase content without altering phase composition shown in Fig. 7. These findings highlight the potential of ultrasonic-assisted CMT + P welding in improving the quality and reliability of Ti-6Al-4V welds for critical applications.

Zhe Sun. [164] study has focused on understanding the details of droplet transfer behaviors during cold metal transfer (CMT) welding of Ti-6Al-4V alloy, which is a crucial material in the aero engine industry due to its exceptional mechanical properties. Unlike typical CMT cycles, the research discovered distinct current and voltage waveforms characterized by a significant current pulse during the short-circuit phase of welding. This pulse plays a vital role in adjusting energy distribution, which helps in facilitating a smoother droplet transfer process. The frequency and size of droplet transfer are found to be influenced by parameters such as wire feed rate and inductance correction, with larger droplets observed at higher feed rates or lower inductance values and result shown in Fig. 8.

Analysis of electrical waveforms and high-speed images explains how these parameters impact droplet behavior shown in Fig. 9. Experimental validation confirms the importance of the current pulse in enhancing energy efficiency, where it contributes a larger proportion of total energy input compared to traditional CMT processes.

This comprehensive understanding not only improves welding stability and quality but also provides essential insights for optimizing CMT techniques in aerospace applications of Ti-6Al-4V alloy, highlighting its relevance for advancing industrial welding practices.

Fuyang Gao [165] Recent studies have demonstrated that Cold Metal Transfer (CMT) welding is an effective technique for joining Ti-6Al-4V titanium alloys due to its low heat input and stable droplet transfer characteristics. Titanium alloys are widely used in aerospace and marine structures because of their high specific strength and excellent corrosion resistance. However, their high melting point, strong chemical reactivity, and low thermal conductivity make welding difficult and may

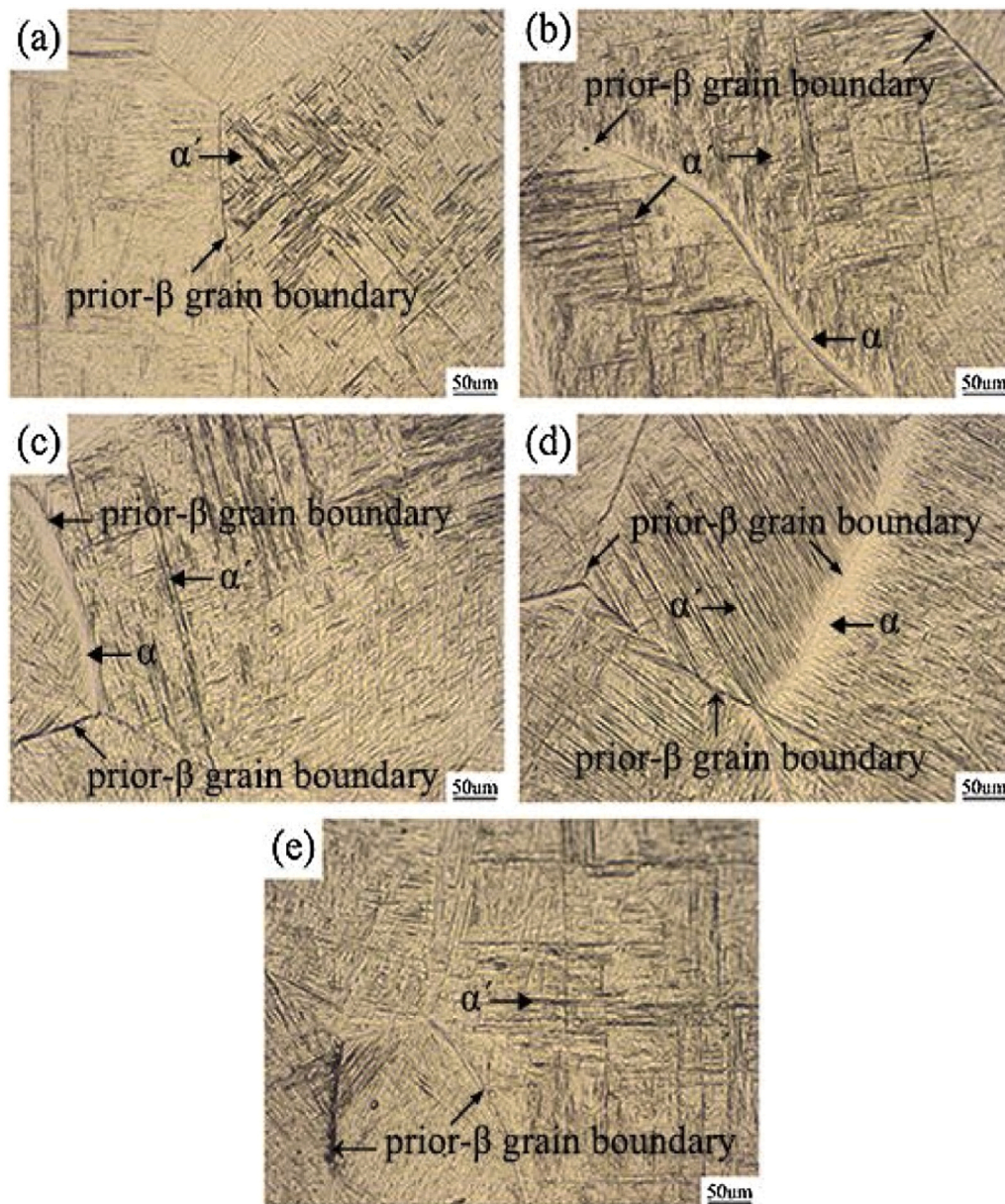


Fig. 7. The phases present in the weld zone Microstructure (α' martensite, α phase, and retained β phase) (a) sample 1; (b) sample 2; (c) sample 3; (d) sample 4; (e) sample 5 [163].

lead to grain coarsening and reduced joint performance. Conventional welding processes such as TIG often produce high heat input and deformation, while electron beam welding requires vacuum conditions that limit industrial applications. CMT welding overcomes several of these limitations by combining the advantages of MIG/MAG and TIG welding while maintaining lower heat input and minimal spatter. Research indicates that the weld metal formed during CMT welding mainly consists of acicular α' martensite, massive α , and Widmanstätten $\alpha+\beta$ microstructures. The fusion zone generally contains fine martensitic structures, whereas the heat-affected zone exhibits acicular martensite with residual β phase. Microhardness measurements show the highest hardness in the fusion zone, followed by a gradual decrease toward the heat-affected zone and base metal. Mechanical testing has shown that the tensile strength of CMT welded joints can reach approximately 930 MPa, which is comparable to that of the base metal. In addition, the fracture during tensile testing typically occurs in the base metal rather than in the weld region, indicating good joint integrity. Impact toughness of the weld metal has also been reported to be about 33 % higher

than that of the base metal due to the presence of acicular microstructures that hinder crack propagation. Overall, these findings demonstrate that CMT and CMT+P welding processes can produce high-quality titanium alloy joints with improved mechanical performance and reduced welding defects.

3.2. MIG welding

W Nishikawa [166] study has focused on the Pulsed Metal Inert Gas (MIG) welding of titanium highlights the critical role of accurate and comprehensive information for optimizing welding parameters and ensuring weld quality. Studies emphasize the benefits of adding oxidizing gases, such as oxygen and carbon dioxide, to the argon shielding gas to stabilize the arc and improve droplet transfer. Various researchers have investigated the effects of different concentrations of these gases, observing significant impacts on mechanical properties and arc stability. Experimental setups typically involve varying the oxygen content between 1 % and 5 % and carbon dioxide content between 2 %

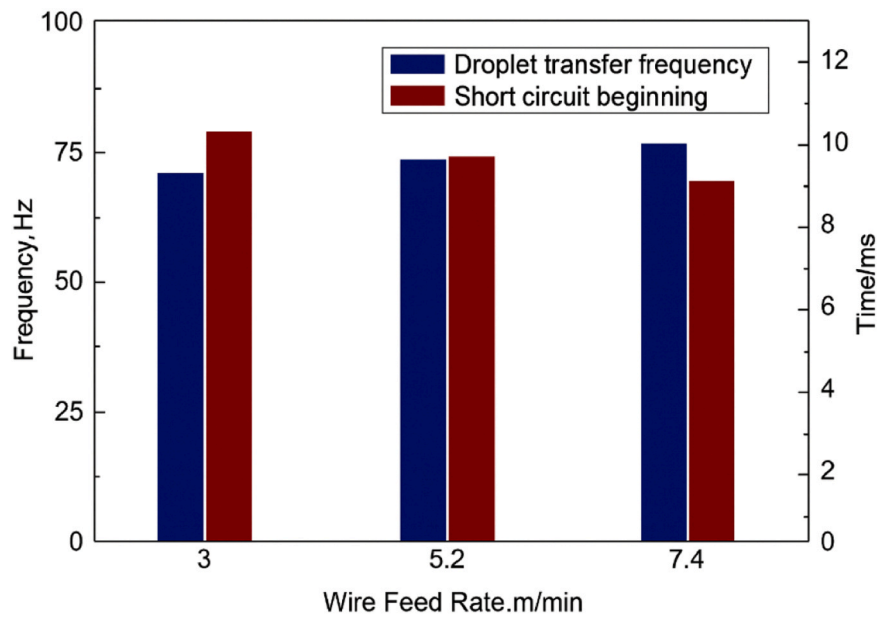


Fig. 8. Relationship between wire feed rate and droplet transfer behavior [164].

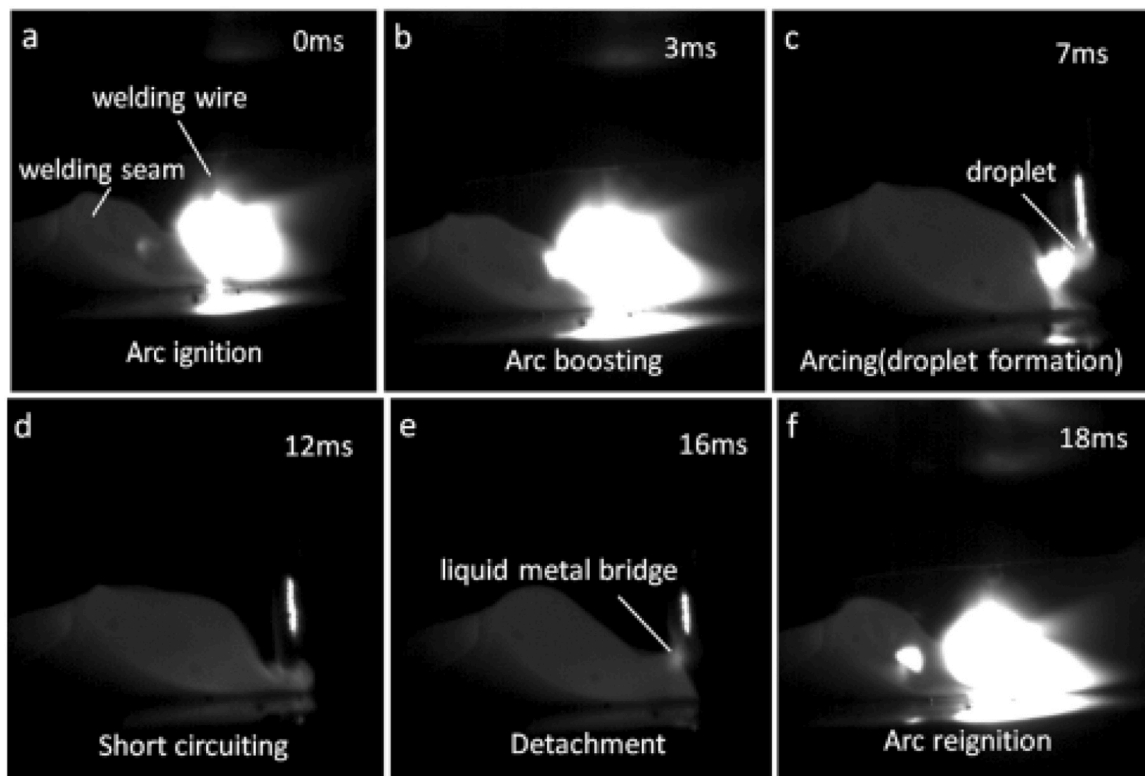


Fig. 9. High-speed images of droplet transfer in one Ti-6Al-4V alloy CMT cycle [164].

and 10 %, with subsequent evaluations of the weld quality through mechanical testing, including Vickers hardness and tensile tests. Standards such as WES 7102 and JIS H 4600 are frequently used as benchmarks. Findings consistently demonstrate that small additions of oxidizing gases can enhance the tensile strength and hardness distribution of titanium welds shown in Fig. 10, provided the oxygen content is kept below critical thresholds (e.g., 2 %).

Furthermore, fracture analysis using Scanning Electron Microscopy (SEM) reveals a transition from ductile to brittle fractures with

increasing oxidizing gas content, underscoring the delicate balance required in gas composition. Overall, the consensus in the literature underscores the potential of pulsed MIG welding, with optimized gas compositions, to achieve stable and high-quality titanium welds.

Chuanchuan Jia [167] study to novel approach was investigated to enhance MIG welding of Ti-6Al-4V alloy by introducing a controllable coaxial dual-path shielding gas-flow field. Conventional MIG welding of titanium alloys often suffers from arc instability, spatter formation, and inadequate shielding, which can reduce weld quality. These problems

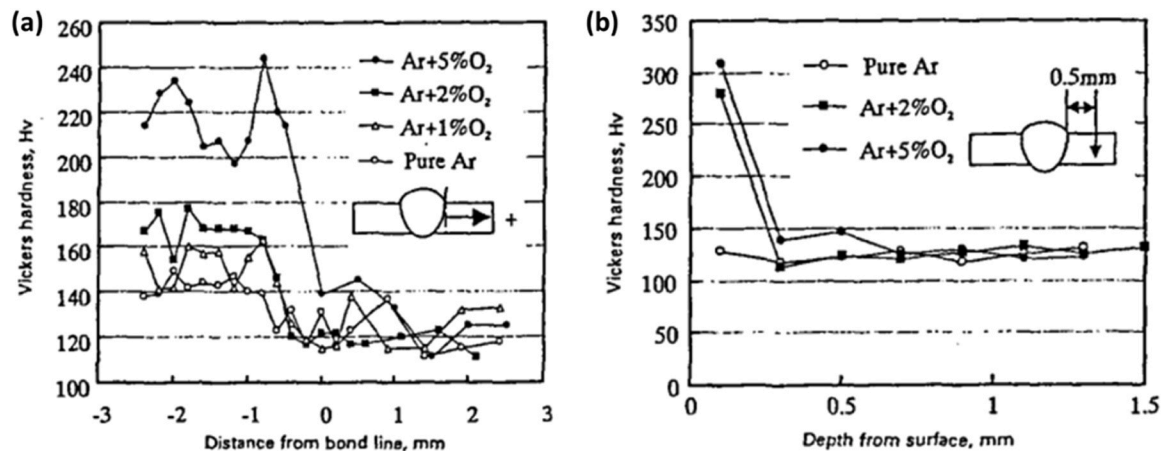


Fig. 10. (a) Hardness distribution of welds (horizontal direction), (b) Hardness distribution of welds (perpendicular direction) [166].

are critical because titanium alloys are highly reactive at elevated temperatures, and exposure of the molten pool to air can cause oxidation and embrittlement. To address these limitations, a dual-path shielding gas system was designed to regulate arc behavior and droplet transfer during welding. The experimental setup incorporated a custom composite welding torch, MIG power source, and wire feeder, while high-speed imaging and electrical signal acquisition were used to monitor droplet transfer and arc characteristics. Numerical simulations were also conducted to analyze arc plasma behavior. Results showed that the combined internal and external gas flow compressed the arc in both longitudinal and transverse directions, increasing arc energy density and penetration depth. The system improved the short-circuit transfer mode by extending the voltage range, reducing peak current, and shortening short-circuit duration. In globular transfer mode, the upper welding current limit decreased while droplet transfer frequency increased, resulting in more stable transfer and lower heat input. Improved penetration and narrower weld beads were also observed. However, interaction between the gas flows occasionally caused turbulence that introduced pores into the weld pool. Overall, the dual-path shielding gas-flow field enhanced welding stability and process flexibility, leading to improved weld performance in Ti-6Al-4V MIG welding.

Xi Niu [168] study MIG and hybrid welding technologies have shown promising developments for joining titanium and its alloys, several limitations still exist. Conventional MIG welding often encounters difficulties when welding medium and thick plates due to uneven heat distribution across the weld surface. The high surface tension and poor wettability of molten titanium, combined with its low thermal conductivity, can lead to heat accumulation on the welding wire, resulting in arc instability, spatter, porosity, and other weld defects. Because of these challenges, standalone MIG welding is rarely applied to titanium alloys. Advanced techniques such as pulsed MIG, PMC MIG, and CMT improve weld stability and heat input control, but they require sophisticated equipment, precise parameter control, and high shielding gas purity. Hybrid processes such as laser-MIG and plasma-MIG welding further enhance penetration and productivity; however, they involve complex system integration, expensive equipment, and strict process control. In addition, these high-energy processes may generate significant thermal stresses that increase the risk of cracking if parameters or shielding conditions are not properly optimized. The large number of adjustable parameters also makes process optimization challenging and requires high technical expertise. Moreover, the development of specialized filler materials for titanium welding remains limited, which restricts further improvements in weld performance. Therefore, continued research is necessary to simplify process control, reduce costs, and develop improved welding consumables for wider industrial application.

Huijing Zhang [169] Recent studies have investigated the use of Metal Inert Gas (MIG) welding for the repair of TC4 (Ti-6Al-4V) titanium

alloy components widely used in aerospace structures. Titanium alloys often develop defects such as cracks, wear, and surface damage during service, requiring effective repair techniques to restore structural integrity. Compared with conventional repair methods, arc-based wire repair technologies such as MIG welding offer higher deposition efficiency, better penetration capability, and lower processing costs. Research comparing conventional MIG (C-MIG) and pulsed MIG (P-MIG) modes shows that the weld center mainly consists of β phase and acicular α' martensite formed due to rapid cooling during welding. The P-MIG process enhances molten pool stirring and droplet transfer, resulting in finer martensitic structures compared with conventional MIG welding. Grain refinement significantly improves microhardness and tensile strength in the repaired region. Experimental results indicate that the weld center exhibits the highest hardness due to the presence of fine α' martensite and dense phase boundaries. Moreover, P-MIG repaired specimens demonstrate higher tensile strength than C-MIG repaired specimens. Fracture analysis reveals a mixed ductile-brittle fracture behavior characterized by cleavage steps, tearing ridges, and dimples. Overall, pulsed MIG welding shows strong potential for improving the microstructure and mechanical performance of repaired TC4 titanium alloy components.

Guangfu Wu [170] Recent studies have investigated the influence of ultrasonic impact treatment (UIT) on the microstructure and mechanical performance of MIG-welded TC4ELI titanium alloy joints. TC4ELI alloy is widely used in aerospace and marine structures because of its high strength, fracture toughness, and corrosion resistance. However, MIG welding of titanium alloys often produces coarse microstructures and high residual tensile stresses due to uneven thermal cycles during welding. Researchers have shown that the weld zone mainly consists of β -phase, acicular α' martensite, and acicular α phase, while the heat-affected zone contains interlaced martensitic structures formed during rapid cooling. Application of ultrasonic impact treatment introduces high-frequency vibration energy that causes plastic deformation and promotes grain refinement through dislocation multiplication and subgrain boundary formation. This treatment significantly modifies the residual stress state, converting harmful tensile stresses in the weld region into beneficial compressive stresses. Experimental results indicate that UIT increases the weld center microhardness by about 9.5 % and improves impact toughness of the weld zone and heat-affected zone by approximately 17 % and 23 %, respectively. However, the influence of UIT on tensile strength is relatively small, with only a slight improvement observed. Overall, ultrasonic impact treatment effectively refines microstructure, reduces residual stresses, and enhances the toughness of MIG-welded TC4ELI titanium alloy joints, demonstrating its potential for improving the performance and reliability of titanium alloy welded structures.

While the addition of oxidizing gases improves arc stability and

mechanical performance, their use must be carefully controlled not only to prevent embrittlement but also to minimize environmental impact associated with increased oxidation and gas emissions.

3.3. Laser-MIG welding

Ruifeng Li [171] study has focused on the Titanium alloys, particularly Ti-Al-Zr-Fe, are essential in aerospace and biomedical industries due to their high strength-to-weight ratio and excellent corrosion resistance. However, welding these alloys poses challenges, such as oxidation and cracking. Recent studies have focused on advanced welding techniques to overcome these issues. Laser beam welding (LBW) and laser-MIG hybrid welding (LAMIG) are two methods gaining attention for their potential to produce defect-free joints. LBW is known for its high precision and low heat input, minimizing thermal distortion. However, it often results in high hardness in the fusion zone, which can reduce ductility. Conversely, LAMIG, which combines laser and metal inert gas welding, offers higher heat input and the addition of filler material, enhancing weld formation and reducing hardness in the fusion zone. Recent research demonstrated that both LBW and LAMIG can achieve high-speed welding (1.8 m/min) and maintain weld integrity without defects like oxidation or porosity. Notably, LAMIG-welded joints exhibited superior mechanical properties, including higher ductility and better strength-ductility combinations, compared to LBW shown in Table 6. These findings suggest that LAMIG is more advantageous for welding Ti-Al-Zr-Fe alloys, providing a feasible method for producing high-quality, strong, and ductile titanium joints.

Shuling Chen [172] study has focused on the Porosity is a critical issue in laser-MIG hybrid welding of titanium alloys, affecting the quality and strength of the weld. Recent studies have concentrated on the impact of the distance at which the laser is out of focus on the formation of porosity in joints made of TA2 titanium alloy. Both experimental and numerical methods, which included using a volume of fluid (VOF) model in FLUENT, were employed to examine how the laser-induced keyhole, MIG droplet, and melt pool interacted. The results indicate that increasing the defocusing distance from +6 mm to +12 mm significantly decreases porosity shown in Fig. 11.

This decrease is ascribed to the reduction in laser energy density, which creates a shallower keyhole and weaker liquid metal vortex flow, allowing the surrounding liquid metal to fill the space left by the collapsing keyhole. Experimental observations have confirmed that a greater defocusing distance results in fewer pores, and simulations have shown a corresponding reduction in keyhole depth and pore size shown in the Fig. 12.

Furthermore, a larger defocusing distance enhances weld stability and appearance, resulting in a wider weld while still maintaining consistent penetration depth. These findings highlight the importance of optimizing the defocusing distance to improve the quality of welds and reduce defects in the laser-MIG hybrid welding of titanium alloys, providing valuable insights for refining industrial welding processes.

Zhenxing He [173] study has focused on the Porosity is a significant concern in laser-MIG hybrid welding of titanium alloys as it impacts the mechanical properties and overall quality of the welded joints. Recent studies on TC4 titanium alloy with a thickness of 5 mm have explored the relationship between defocusing distance and porosity formation. Using high-speed cameras to observe the keyhole evolution, researchers found that keyhole stability is compromised at defocusing amounts of

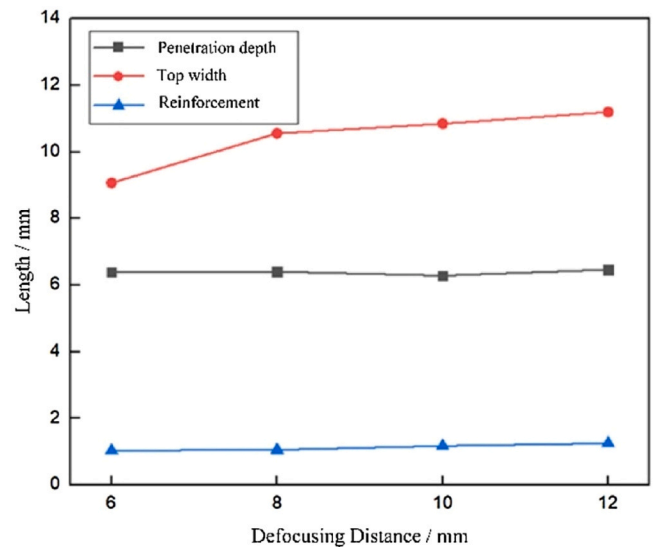


Fig. 11. The macro-morphology of welds surface obtained with different defocusing distances [172].

+2, 0, -2, and -4 mm, with the highest porosity (4.21 %) occurring at -2 mm. The keyhole only collapses at the root, leading to process porosity. Remarkably, adjusting the defocusing amount to +4 and -6 mm significantly reduces porosity by over 80 %, and no porosity is observed at +6 mm. This adjustment improves keyhole stability, preventing narrow and distorted shapes. The study highlights the critical role of keyhole dynamics in porosity formation and demonstrates that process porosity can be effectively mitigated by optimizing defocusing distances. The findings suggest that a stable keyhole is essential for minimizing porosity and enhancing the weld quality of TC4 titanium alloy joints. These insights are valuable for improving laser-MIG hybrid welding processes, emphasizing the need for precise control of defocusing parameters to achieve superior weld integrity.

Cui Li [174] study has focused on the Commercially pure titanium (CP-Ti) is commonly used in industries that require high strength and corrosion resistance. Therefore, it is crucial to optimize welding parameters to ensure the integrity and performance of welded joints. Recent research has focused on studying the effects of various parameters in fiber laser-metal inert gas (MIG) hybrid welding of CP-Ti sheets. Key parameters such as MIG welding current, welding speed, laser-arc distance, and defocused distance of the laser beam were systematically varied to assess their impact on bead appearance, weld shape shown in Fig. 13, and tensile properties, particularly elongation and ultimate tensile strength (UTS).

Through experiments using a 2 kW fiber laser in an argon-filled environment to prevent contamination, it was discovered that hybrid welding produces more uniform welds compared to MIG or fiber laser welding alone. The study revealed that while the UTS of the welded joints were generally unaffected by the welding parameters due to higher microhardness in the fusion zone, elongation was significantly influenced by the process conditions. Optimal parameters were found to enhance elongation. All hybrid welded joints fractured in the base metal, indicating superior strength in the weld metal shown in Fig. 14.

These findings emphasize the importance of precise control over welding parameters to achieve desirable mechanical properties and structural integrity in CP-Ti welds. The research also points to the need for further exploration into the relationship between welding parameters, microstructure, and mechanical properties to optimize welding procedures and structural designs for CP-Ti applications.

Chen Yue [175] study has focused on the impact of transverse magnetic fields on arc stability and droplet transfer in laser-MIG hybrid welding of Ti-6Al-4V titanium alloy has been a subject of considerable

Table 6

Tensile and bending test results of parent metal, LBW and LAMIG welded joints.

Welding Process	Tensile test		Bending test	
	σ_b (MPa)	Fracture location	Face bending angle (°)	Root bending angle (°)
Parent metal	690	-	>120	-
LBW	696	Parent metal	90 (fusion line)	>120
LAMIG	709	Parent metal	>120	>120

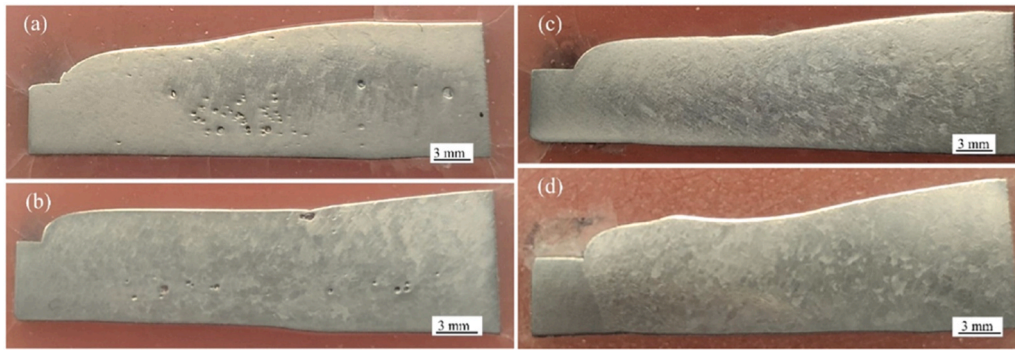


Fig. 12. Macro-morphology of the longitudinal cross-section of the laser-MIG hybrid welding joint with different defocusing distance: a) + 6 mm; b) + 8 mm; c) + 10 mm and d) + 12 mm [172].

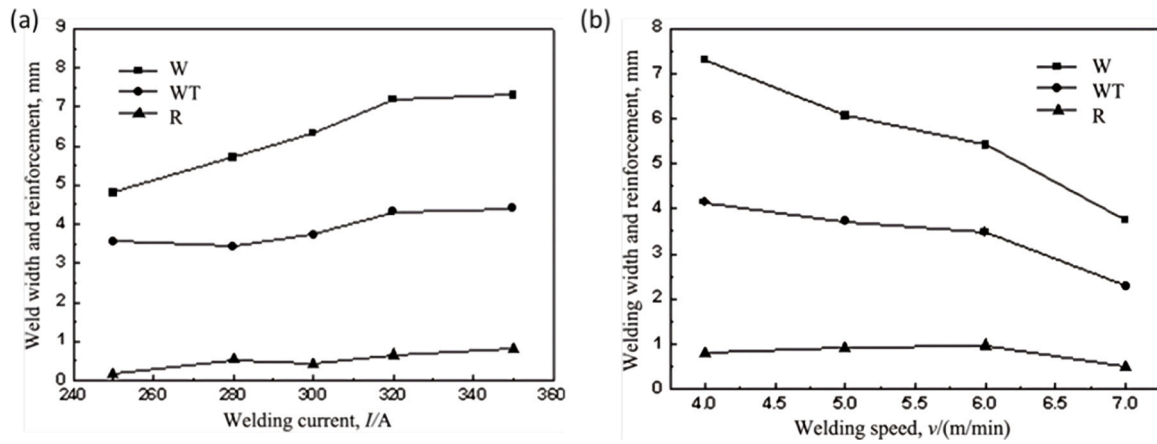


Fig. 13. (a) Effect of welding current on the weld width and reinforcement height. (b) Effect of welding speed on the weld width and reinforcement height [174].

research. Recent studies have examined the effects of magnetic field strengths ranging from 0 mT to 32 mT on the welding process. Key findings suggest that an optimal magnetic field enhances arc stability, droplet detachment efficiency, and reduces welding defects such as incomplete fusion and undercutting. Specifically, a 24 mT magnetic field significantly improves welding performance by reducing the maximum arc area by 48.7 % and decreasing the variance from 2.81 to 1.06 mm². This field strength promotes a transition from short-circuit to meso-spray droplet transfer, enhancing detachment efficiency and minimizing contact time with the molten pool. The study involved a 4-mm-thick Ti-6Al-4V base metal and a 1-mm-thick filler wire, with surface preparation involving acetone cleaning and laser oxidation removal. Image processing techniques were used to analyze arc characteristics and droplet transfer behavior. The research concluded that while minimal magnetic fields (8 mT) have negligible effects, excessive fields (32 mT) destabilize the welding process. The optimal magnetic field of 24 mT significantly improves arc stability and droplet transfer efficiency, underscoring the importance of magnetic field control in enhancing the quality and reliability of welds in titanium alloys in shown in Fig. 15. These findings offer valuable insights for optimizing welding parameters to achieve superior weld integrity in Ti-6Al-4V applications.

Anguo Huang [176] study has focused on the Groove depth plays a critical role in influencing plasma behaviors and weld quality in laser-MIG hybrid welding, especially for joining medium and thick plates. Recent studies have focused on the effects of varying groove depths on plasma temperature and electron density, addressing a gap in understanding how groove configurations impact welding outcomes. Experiments, utilizing stair-stepping plates with groove depths ranging from 0 to 20 mm, combined with an IPG fiber laser and a FRONIUS

digital welder, revealed significant changes in plasma characteristics. The use of a high-speed camera and spectral acquisition system provided detailed insights into the welding process. Results indicated that increasing groove depth led to a decrease in plasma temperature from 10,589 K to 8527 K, while electron density increased from $1.30 \times 10^{16} \text{ cm}^{-3}$ to $1.44 \times 10^{16} \text{ cm}^{-3}$ shown in Fig. 16. Deeper grooves resulted in more laser-induced metal vapor, affecting heat convection and radiation, thus lowering plasma temperatures.

Additionally, increased groove constraint space led to a decrease in plasma profile and droplet transfer frequency, while enhancing weld penetration depth and contracting the arc column, reducing weld width from 6.7 mm to 4.5 mm shown in Fig. 17.

These findings highlight the significant influence of groove depth on plasma behavior and weld quality, suggesting that optimizing groove configurations can improve welding processes for thicker plates. The study provides a foundation for future research aimed at refining welding technologies and enhancing the structural integrity of welded joints in industrial applications.

R. Spina [177] study has focused on the hybrid Laser-GMAW welding of Ti-6Al-4V titanium alloy sheets has been extensively researched due to its ability to combine the strengths of laser welding and gas metal arc welding (GMAW). This method utilizes the precise, high-energy density of laser welding to create keyhole formation and bridge gaps, along with the adaptable heat input of GMAW to address issues such as high thermal gradients and low mechanical resistance. Recent research on 3 mm thick Ti-6Al-4V sheets, using a double fillet welded T-joint geometry, has shown significant improvements in bead morphology, microstructure, and mechanical properties, especially micro-hardness, when compared to base metals. The experimental setup involved a 6 kW CO₂ laser and a GMAW machine working together, where the laser formed

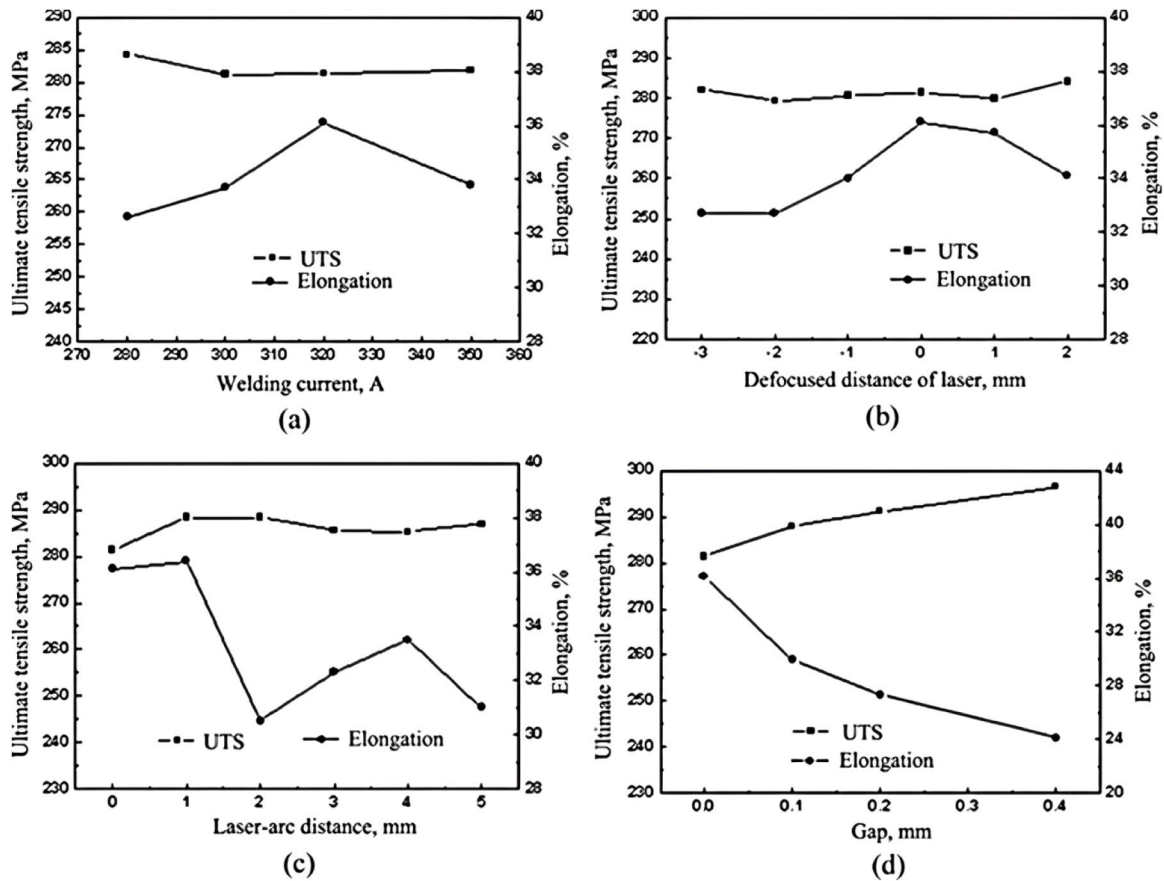


Fig. 14. Effect of welding parameters on tensile properties of the welded joints: (a) welding current; (b) defocused distance; (c) laser-arc distance; and (d) gap [174].

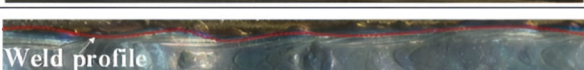
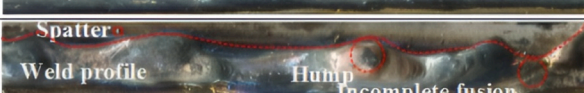
External magnetic field	Weld appearance
0 mT	
8 mT	
16 mT	
24 mT	
32 mT	

Fig. 15. The weld appearance under different external magnetic field [175].

the root pass, and GMAW completed the weld. Results indicated that hybrid welding met ANSI/AWS standards, with the 1 F position (fillet weld in the flat position) being optimal for titanium alloys. Increased welding speeds were found to decrease the fusion zone area, under-fill, and heat-affected zone width. Despite some issues with reduced ductility due to micropores and aluminum oxides, hybrid welded joints displayed comparable or slightly superior joint strength. Future research is

suggested to explore new beam sources with high output power, such as fiber or disk lasers, to enhance penetration depth and process reliability by reducing plasma plume formation. These findings highlight the potential of hybrid welding techniques to enhance the quality and performance of welded titanium alloys, making them highly relevant for industrial applications. Table 7 summarizes the welding parameters (current, voltage, wire feed speed, welding speed, gas flow rate) and the

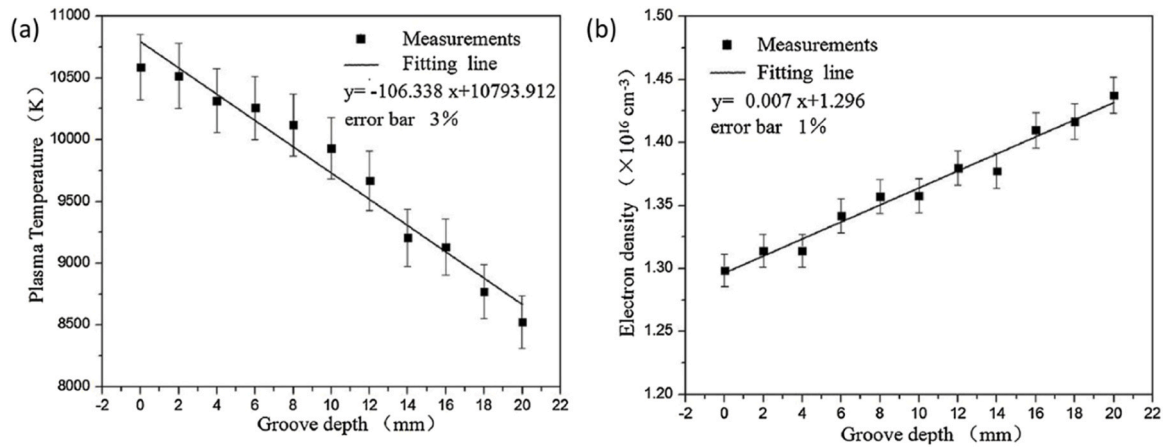


Fig. 16. (a) The average temperature of plasma at different groove depths., (b) Average electron density at different groove depths [176].

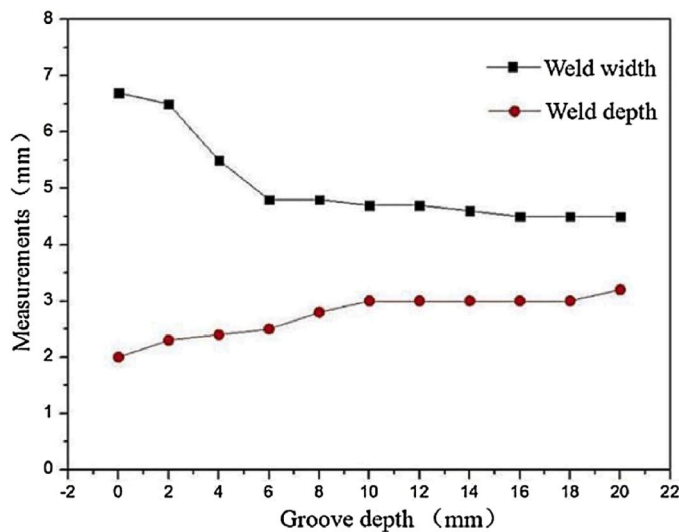


Fig. 17. Weld width and penetration depth of weld at different groove depths [176].

corresponding tensile strength and failure locations reported for different titanium alloys joined using CMT, MIG, and laser-MIG hybrid welding processes across varying material thicknesses and joint configurations.

Chuang Cai [178] study evaluates the stability of narrow-gap oscillating laser-MIG hybrid welding (NG-LAHW) for Ti-6Al-4V titanium alloy by analyzing the influence of different laser oscillation patterns on weld quality and porosity formation. Experiments were conducted on 16 mm thick Ti-6Al-4V plates using ERTi-6Al-4V filler wire and a disk laser combined with pulse MIG welding. Four oscillation paths linear, circular, 8-shaped, and infinity were investigated to understand their effect on welding behavior. High-speed imaging, keyhole observation, X-ray inspection, and metallographic analysis were employed to monitor arc behavior, molten pool dynamics, and defect formation. Results showed that linear, 8-shaped, and infinity patterns produced convex weld surfaces with irregular morphology. In contrast, circular oscillation generated a smooth concave weld with improved surface quality. The circular pattern also achieved the lowest porosity level (0.16 %), representing a significant reduction compared with the linear pattern. Improved weld quality was attributed to more uniform laser energy distribution and enhanced keyhole stability. Circular oscillation minimized keyhole fluctuations and prevented arc deflection, resulting in stable arc laser interaction. It also increased droplet transfer

frequency while reducing droplet impact force, promoting smoother molten pool flow. Consequently, circular laser oscillation significantly improves process stability and weld integrity in NG-LAHW of titanium alloys.

To comprehensively evaluate the recent advances in welding of titanium alloys, various studies have been reviewed, focusing on Cold Metal Transfer (CMT), Metal Inert Gas (MIG), and hybrid Laser-MIG (LAMIG) techniques. Each method addresses specific challenges associated with titanium welding, such as grain coarsening, porosity formation, arc instability, and reduced ductility in the fusion zone. Researchers have proposed innovative modifications, including ultrasonic vibration, optimized shielding gas composition, defocusing distance adjustments, magnetic field assistance, and groove design strategies to enhance weld quality and reliability. The summarized findings highlight how these innovations influence microstructure, droplet transfer behavior, porosity control, and mechanical performance, offering valuable insights into the optimization of welding processes for high-performance titanium applications. To further improve the clarity of comparison among different welding approaches, Table 8 presents a structured overview of the key challenges, modification strategies, and resulting improvements in weld quality reported in recent studies. This comparison highlights how process modifications such as ultrasonic assistance, magnetic field application, and parameter optimization influence weld microstructure and mechanical properties.

4. Comparison and discussion of processes

Recent studies published in the last few years have increasingly emphasized the role of advanced process control and hybrid welding techniques in improving the weldability of titanium alloys. Compared with traditional arc welding methods, modern techniques such as CMT and Laser-MIG hybrid welding provide better control of heat input, droplet transfer dynamics, and melt pool stability. These advancements contribute to refined microstructures, reduced residual stresses, and improved mechanical properties in welded joints. Consequently, contemporary research has shifted toward optimizing hybrid energy sources and integrating auxiliary process assistance to achieve defect-free titanium welds suitable for aerospace and biomedical applications.

A thorough understanding of how process parameters affect heat input, droplet transfer, and resulting microstructures can be gained by comparing different welding methods for titanium and its alloys.

In the CMT process, a synergistic control between the current waveform and wire retraction leads to an extremely low heat input (10–15 % of conventional MIG). The short-circuit and wire-pullback cycle that happens every now and then cuts down on spatter and makes it easier to control the detachment of droplets. This process makes α' martensitic microstructures finer and lowers residual stresses, which

Table 7
summarizes the welding parameters reported in the literature, including current, voltage, wire feed speed, welding speed, shielding gas flow rate, and corresponding tensile strength values for different titanium alloys joined using CMT, MIG, and Laser-MIG hybrid welding processes. This unified dataset allows direct comparison of process parameters and mechanical performance across different welding techniques.

Sr. No	Ref.	Welding Process	Material with thickness	Joining process	Current (A)	Voltage (V)	Wire feed speed	Welding speed	Gas flow rate	tensile strength (MPa)	tensile failure location
1	[163]	CMT + P welding	Ti-6Al-4V (1.2 mm)	Butt joint	540-70	560-70	5 m/min	0.24 m/min	15 L/min	974.2-1052.2	Weld Zone
3	[164]	CMT	Ti-6Al-4V (8 mm)	-	40-150	13-16	2-9 m/min	0.39 m/min	15 L/min	-	-
5	[166]	MIG	Titanium (1.5 mm)	Butt joint	110	20	5.4 m/min	0.54 m/min	15 L/min	270-410	Base metal surface side
6	[171]	Laser MIG	Ti-Al-Zr-Fe titanium alloy (4 mm)	Butt joint	-	30	10 m/min	1.8 m/min	25 L/min	709	base metal
7	[172]	Laser MIG hybrid	CP-Titanium (6 mm)	Butt joint	120	19.1	-	0.48 m/min	-	-	-
8	[173]	Laser MIG hybrid	Ti-6Al-4V (5 mm)	Butt joint	120	-	-	0.79 m/min	17 L/min	-	-
9	[174]	Laser MIG hybrid	CP-Titanium (1.5 mm)	Butt joint	250-350	-	-	3.99 m/min	15,20 L/min	280-290	Base metal
10	[175]	Laser MIG hybrid	Ti-6Al-4V (4 mm)	Butt joint	160	-	-	1.99 m/min	-	-	-
11	[176]	Laser MIG hybrid	Ti-6Al-4V (4 mm)	Butt joint	120	15.6	-	0.6 m/min	25 L/min	-	-
12	[177]	Laser MIG	Ti-6Al-4V (3 mm)	T joint	580	40	5 m/min	1.99 m/min	-	-	-

* Blank entries in Table 7 indicate parameters that were not explicitly reported in the corresponding literature sources.

Table 8

Comparative Summary of Welding Techniques for Titanium Alloys.

Welding Process	Key Challenge	Approach / Modification	Outcome
CMT + P (Ultrasonic-assisted) [163]	Columnar grain growth, poor ductility	Ultrasonic vibration via tungsten needle	Refined grains, higher UTS, mixed ductile + brittle fracture, improved weld reliability
CMT (Droplet Transfer) [164]	Unstable droplet transfer, inefficient energy use	Control of current pulse, wire feed rate, inductance	Stable droplet transfer, improved energy efficiency, better weld quality
MIG (with oxidizing gas) [166]	Arc instability, poor droplet transfer	Addition of O ₂ (≤2 %) or CO ₂ (2-10 %) to argon	Stable arc, better droplet transfer, improved tensile strength & hardness; excess gas → brittle fracture
LBW (Laser Beam Welding) [171]	High hardness in fusion zone, reduced ductility	Precision low heat input laser	Defect-free welds, but reduced ductility due to hardness
LAMIG (Laser-MIG Hybrid) [172]	Balancing strength & ductility	Combined laser + filler addition	Higher ductility, improved strength-ductility, defect-free welds
LAMIG (Defocusing Distance) [173]	Porosity due to unstable keyhole	Adjust defocusing distance (+6 to +12 mm optimal)	>80 % porosity reduction, stable welds, wider bead, improved weld integrity
LAMIG (Parameter Optimization) [174]	Inconsistent mechanical properties	Control current, speed, arc-laser distance	Uniform welds, elongation improved, weld stronger than BM
LAMIG (Magnetic Field) [175]	Arc instability, inefficient droplet transfer	Apply transverse magnetic field (optimal 24 mT)	Arc stability Increase, droplet detachment improved, reduced defects
LAMIG (Groove Depth) [176]	Plasma instability, weld penetration control	Vary groove depth (0-20 mm)	Plasma temp decrease, penetration Increase, weld width decrease, optimized plasma-weld interaction
Hybrid Laser-GMAW [177]	Thermal gradients, low ductility	Combine CO ₂ laser (keyhole) + GMAW (fill)	Improved bead, microstructure & hardness, comparable strength; ductility loss due to micropores/oxides

makes Ti-6Al-4V joints stronger and more flexible. According to Zhao et al. [163], adding pulsing (CMT+P) or ultrasonic vibration makes the grains even finer and stops them from forming columns.

In addition to fusion zone characteristics, the heat-affected zone (HAZ) plays a decisive role in determining weld integrity in titanium alloys. The extent and microstructural evolution within the HAZ are strongly governed by the thermal cycle associated with each welding process. In Cold Metal Transfer (CMT) welding, the significantly reduced heat input results in a narrow HAZ with limited grain growth and refined α' martensitic structures, thereby preserving mechanical properties and minimizing distortion. In contrast, conventional MIG welding introduces higher heat input, leading to a wider HAZ and coarser α -lamellae, which can adversely affect ductility and fatigue resistance. Laser-MIG hybrid welding offers a balanced thermal profile, where the concentrated laser energy promotes rapid heating and cooling, resulting in a comparatively narrow HAZ with refined microstructures. This controlled HAZ evolution contributes to improved strength-ductility combinations and reduced thermal distortion. Therefore, optimization of process parameters is essential to control HAZ characteristics and ensure the structural reliability of welded titanium components [179-181].

MIG welding is strong and can be used in many situations, but it usually has a higher heat input and droplet momentum. Research conducted by Nishikawa et al. [166] indicates that the incorporation of regulated oxygen or CO₂ into argon shielding gas enhances arc stability and wetting characteristics, provided that concentrations remain beneath critical thresholds to prevent embrittlement. Compared to CMT, MIG welding makes a wider fusion zone and coarser α -lamellae, which makes the material less ductile. Even with these problems, MIG is still a good choice for structural titanium parts where accuracy isn't as important because it's easy to use and has a high deposition rate.

A direct quantitative comparison between ultrasonic-assisted CMT + P welding and pulsed MIG welding highlights significant differences in mechanical performance. Zhao et al. [182] reported that ultrasonic-assisted CMT + P welding of Ti-6Al-4V achieved an ultimate tensile strength (UTS) in the range of 974–1052 MPa, which is comparable to or slightly higher than the base metal strength. In contrast, pulsed MIG welding reported by Nishikawa et al. [183] exhibited significantly lower UTS values ranging from 270 to 410 MPa under varying shielding gas compositions. From a ductility perspective, ultrasonic-assisted CMT + P welding demonstrated a refined microstructure dominated by fine α' martensite and equiaxed grains, resulting in a combination of ductile and brittle fracture modes. Conversely, pulsed MIG welding showed a strong dependence on shielding gas composition, where increasing oxygen content improved strength but reduced ductility, leading to a transition toward brittle fracture behavior. It should be noted that direct quantitative comparison is influenced by differences in experimental conditions such as material thickness, welding parameters, and shielding environments. Therefore, while absolute values differ, the observed trends consistently indicate that ultrasonic-assisted CMT + P welding provides superior strength-ductility balance compared to pulsed MIG welding due to its lower heat input and enhanced microstructural refinement. Table 9 presents a Comparative Mechanical Performance of Ultrasonic-Assisted CMT + P and Pulsed MIG Welding for Ti-6Al-4V.

The Laser-MIG hybrid (LAMIG) process is a middle ground that combines the best features of both laser and arc energy sources. The laser beam goes deep into the material with very little heat-affected zone (HAZ), and the MIG arc adds filler metal to make up for gaps in the joint. Li et al. [171] and Chen et al. [172] conducted experiments that revealed that LAMIG welds possess smoother bead profiles, reduced porosity, and enhanced strength-ductility combinations in comparison to standalone MIG or laser beam welding (LBW). Also, changing the defocusing distance from +6 to +12 mm makes keyhole stability better and reduces porosity by more than 80 % [173].

Magnetic field-assisted hybrid welding makes arc stiffness and droplet trajectory control even better, which greatly increases transfer efficiency and cuts down on undercutting [175]. The same goes for optimising the shape of the groove, which is important for heat transfer and plasma behaviour. Increasing groove depth from 0 to 20 mm lowers plasma temperature from about 10,500 K to about 8500 K. This makes it easier for the plasma to penetrate deeper and makes the bead width narrower [176].

Overall, CMT gives you better control and less heat input, MIG gives you productivity and strength, and LAMIG strikes a good balance between strength, ductility, and accuracy. The combination of process optimization, auxiliary fields (magnetic, ultrasonic), and geometric design provides a pathway to achieving defect-free, high-strength welds suitable for aerospace and advanced manufacturing sectors.

4.1. Material compatibility in dissimilar titanium-based joints

The compatibility of materials in dissimilar metal joining is fundamentally governed by differences in melting temperature, thermal conductivity, coefficient of thermal expansion (CTE), and chemical affinity. Titanium exhibits high reactivity and a relatively low thermal conductivity compared to steels and aluminum, which influences heat distribution and solidification behavior during welding [67,184].

Among commonly studied systems, Ti-Al joints are attractive for lightweight applications but suffer from the formation of brittle TiAl₃ intermetallic layers, whose thickness increases with heat input. In contrast, Ti-steel combinations are more challenging due to the formation of Fe-Ti intermetallics (e.g., TiFe, TiFe₂), which are inherently brittle and prone to crack initiation under tensile loading. Nickel-based alloys show relatively better compatibility with titanium due to improved metallurgical affinity and reduced intermetallic brittleness [185,186].

Thermal expansion mismatch is another critical factor influencing joint integrity. During cooling, differential contraction between dissimilar materials generates residual stresses at the interface. These stresses are particularly severe in Ti-Al systems due to the large CTE difference, leading to distortion or interfacial cracking. Similarly, differences in elastic modulus can cause strain incompatibility under service loading conditions [187,188].

Advanced welding techniques such as Cold Metal Transfer (CMT) and Laser-MIG hybrid welding play a significant role in improving compatibility. Their lower and more controlled heat input reduces intermetallic layer thickness and residual stresses. Additionally, the use of interlayers (e.g., Cu, Nb, Ni) and tailored filler materials has been shown to mitigate metallurgical incompatibility by acting as diffusion barriers and stress-relief buffers [189].

Overall, material compatibility in titanium-based dissimilar joints is not solely determined by base material selection but is strongly influenced by process control, thermal management, and metallurgical engineering strategies.

4.2. Comparative discussion of CMT, MIG, and laser-MIG hybrid welding processes

To move beyond a simple compilation of literature, a systematic comparison of Cold Metal Transfer (CMT), Metal Inert Gas (MIG), and Laser-MIG hybrid welding is presented by synthesizing key process-microstructure-property relationships reported in recent studies. These processes differ fundamentally in terms of heat input, droplet transfer mechanisms, weld pool dynamics, and resultant microstructural evolution, all of which ultimately influence the mechanical performance and structural integrity of titanium alloy joints [179,190].

Cold Metal Transfer (CMT) is characterized by a controlled short-circuiting metal transfer mode combined with synchronized wire retraction. This mechanism significantly reduces arc energy input and allows molten droplets to detach at relatively low temperatures. As a result, the process produces a smaller heat-affected zone (HAZ), lower dilution, and improved control over droplet transfer. In titanium alloys such as Ti-6Al-4V, the reduced heat input promotes rapid cooling and leads to the formation of fine α' martensitic structures in the fusion zone. Such microstructural refinement contributes to improved strength while limiting distortion and residual stresses in thin sections [34,191].

In contrast, conventional MIG welding generally involves higher heat input due to continuous arc operation and globular or spray droplet

Table 9
Comparative Mechanical Performance of Ultrasonic-Assisted CMT + P and Pulsed MIG Welding for Ti-6Al-4V.

Process	UTS (MPa)	Ductility Trend	Fracture Mode	Key Mechanism
CMT + P (Ultrasonic)	974–1052 MPa	Moderate-High	Mixed ductile + brittle	Grain refinement, α' martensite
Pulsed MIG	270–410 MPa	Reduced at higher O ₂ content	Ductile → brittle transition	Oxidation, coarse grains

transfer modes. The increased thermal exposure typically results in wider fusion zones and coarser prior- β grains in titanium welds. These microstructural features can reduce ductility and promote defect formation such as porosity or incomplete fusion if shielding conditions are not carefully controlled. Nevertheless, MIG welding remains widely used in industrial applications due to its high deposition rate, operational simplicity, and compatibility with automated welding systems.

Laser-MIG hybrid welding represents a synergistic combination of high-energy laser beam welding and arc welding. The laser component produces deep penetration and a narrow fusion zone through keyhole formation, while the MIG arc supplies filler material and improves joint bridging capability. This hybrid interaction enables improved weld formation, enhanced process stability, and reduced porosity when compared with conventional MIG welding. Furthermore, the steep thermal gradients produced by the laser energy source promote rapid solidification, leading to refined microstructures and improved strength-ductility balance in titanium welds [192,193].

Overall, the comparison highlights that CMT offers superior control over heat input and droplet transfer, MIG provides high productivity and flexibility, while Laser-MIG hybrid welding achieves a balance between penetration capability, mechanical performance, and process efficiency. Understanding these differences is essential for selecting the appropriate welding process for specific titanium alloy applications in aerospace, biomedical, and advanced manufacturing sectors.

CMT produces lower overall heat input and smaller HAZs compared to standard MIG, enabling welding of thinner sections with minimal distortion. The reduced HAZ width also limits grain coarsening and preserves phase stability in Ti-6Al-4V alloys, which is critical for maintaining fatigue resistance and structural integrity.

4.3. Auxiliary control strategies for titanium alloy welding

In recent years, several auxiliary process control strategies have been developed to overcome the inherent challenges associated with titanium alloy welding. Titanium alloys exhibit high chemical reactivity at elevated temperatures and are prone to defects such as porosity, grain coarsening, arc instability, and keyhole collapse. Consequently, researchers have explored various auxiliary techniques including ultrasonic vibration assistance, magnetic field control, shielding gas modification, laser defocusing optimization, and groove geometry design. These approaches aim to regulate weld pool dynamics, improve droplet transfer behavior, stabilize the arc, and refine the resulting microstructure [194,195].

Unlike conventional parameter optimization approaches that primarily adjust current, voltage, and welding speed, auxiliary control methods provide additional physical fields or process modifications that influence the thermal and fluid flow behavior within the weld pool. Such multi-dimensional process control is particularly beneficial for titanium alloys, where small changes in thermal cycles can significantly affect phase transformation and mechanical properties [196–198].

4.3.1. Ultrasonic vibration-assisted welding

Ultrasonic vibration has emerged as an effective auxiliary technique for improving weld pool dynamics and microstructural evolution during titanium alloy welding. The application of high-frequency mechanical vibration into the molten pool promotes enhanced fluid flow and reduces thermal gradients within the weld region. As a result, columnar grain growth is suppressed and equiaxed grain formation is promoted [199,200].

During ultrasonic-assisted Cold Metal Transfer pulse (CMT+P) welding of Ti-6Al-4V alloys, ultrasonic vibration introduced through a tungsten probe generates acoustic cavitation and acoustic streaming within the molten pool. These phenomena disrupt dendritic growth and promote nucleation of fine grains. Consequently, the weld metal exhibits refined microstructures consisting of α' martensite, α phase, and retained β phase [201,202].

Experimental studies have demonstrated that ultrasonic vibration significantly enhances the ultimate tensile strength of welded joints by reducing grain size and improving the uniformity of the microstructure. Furthermore, ultrasonic vibration influences the orientation of grains in the weld zone without altering the chemical composition of the weld metal. These effects lead to improved mechanical properties and reduced susceptibility to cracking [203].

However, several limitations exist. Excessive ultrasonic power may destabilize the molten pool and cause irregular weld bead morphology. Additionally, the implementation of ultrasonic vibration requires specialized equipment and precise positioning of the ultrasonic probe relative to the weld pool.

4.3.2. Magnetic field assisted arc control

Magnetic field assistance is another promising auxiliary technique used to control arc behavior and droplet transfer during titanium welding. When a transverse magnetic field is applied to the welding arc, electromagnetic forces interact with the plasma column, altering the arc shape and stabilizing the droplet transfer process [204].

In laser-MIG hybrid welding of Ti-6Al-4V alloys, the application of a transverse magnetic field has been shown to significantly improve arc stability and droplet detachment efficiency. At an optimal magnetic field intensity of approximately 24 mT, the arc becomes more concentrated and stable, reducing fluctuations in arc area and improving heat distribution within the weld pool.

Magnetic field assistance also facilitates the transition of droplet transfer modes from short-circuit transfer to meso-spray transfer. This improves droplet detachment efficiency and reduces the time during which molten droplets remain attached to the filler wire. As a result, weld defects such as undercutting and incomplete fusion are minimized [205].

Despite these advantages, the effectiveness of magnetic field assistance strongly depends on field intensity. Weak magnetic fields produce negligible effects, while excessive magnetic fields may destabilize the arc and cause irregular droplet transfer. Therefore, precise control of magnetic field strength is essential for achieving optimal welding performance.

4.3.3. Shielding gas composition optimization

Shielding gas composition plays a critical role in titanium welding because titanium has a strong affinity for oxygen, nitrogen, and hydrogen at elevated temperatures. In conventional MIG welding, pure argon is commonly used as a shielding gas. However, small additions of oxidizing gases such as oxygen or carbon dioxide can significantly improve arc stability and droplet transfer behaviour [206].

Studies have shown that the addition of small amounts of oxygen (typically less than 2 %) to argon shielding gas enhances arc conductivity and stabilizes the welding arc. Improved arc stability leads to more uniform droplet transfer and better wetting behavior of the molten metal. As a result, welded joints exhibit improved tensile strength and hardness distribution [207,208].

Nevertheless, excessive oxygen content can lead to embrittlement of the weld metal due to oxygen pickup by titanium. Therefore, careful control of shielding gas composition is necessary to balance arc stability with weld metal integrity.

4.3.4. Laser defocusing and keyhole stability control

In laser-MIG hybrid welding, the stability of the laser-induced keyhole plays a critical role in determining weld quality. Keyhole instability can result in pore formation and irregular weld morphology. One effective approach to controlling keyhole stability is adjusting the defocusing distance of the laser beam [209,210].

Experimental studies indicate that increasing the laser defocusing distance from approximately +6 mm to +12 mm significantly reduces porosity in titanium welds. This reduction occurs because larger defocusing distances decrease the laser energy density, producing a

shallower and more stable keyhole. The resulting molten metal flow becomes less turbulent, allowing the surrounding liquid metal to fill voids created during keyhole collapse [211].

As a consequence, welds produced under optimized defocusing conditions exhibit improved surface morphology, reduced porosity, and enhanced structural integrity. However, excessive defocusing may reduce penetration depth and decrease welding efficiency. Therefore, the defocusing distance must be optimized according to material thickness and welding parameters.

4.3.5. Groove Geometry and Plasma Behavior Control

Groove geometry is another important parameter that influences plasma behavior and heat transfer during hybrid welding processes. In welding of medium and thick titanium plates, groove depth significantly affects plasma temperature, electron density, and weld penetration.

Experimental investigations show that increasing groove depth reduces plasma temperature while increasing electron density within the arc region. These changes affect the droplet transfer frequency and heat distribution in the weld pool. Deeper grooves promote deeper penetration and narrower weld beads, which can improve weld strength and structural stability [132,212,213].

However, excessively deep grooves may reduce droplet transfer frequency and complicate arc stability. Therefore, careful design of groove geometry is necessary to balance penetration depth, weld width, and arc stability.

4.3.6. Limitations and future development of auxiliary techniques

Although auxiliary control methods provide significant improvements in weld quality, several limitations remain. First, most auxiliary techniques require additional equipment and process control systems, increasing the complexity and cost of welding operations. Second, the interaction between multiple auxiliary fields (e.g., ultrasonic vibration and magnetic fields) is not yet fully understood [214–216].

Future research should focus on integrating auxiliary techniques with advanced monitoring technologies such as high-speed imaging, infrared thermography, and machine learning-based process optimization. Such approaches may enable real-time control of weld pool dynamics and further improve the reliability of titanium alloy welding [217,218].

To provide a clearer understanding of the mechanisms and effectiveness of different auxiliary control strategies, Table 10 summarizes the applicable welding processes, fundamental mechanisms, control effects, and limitations of major auxiliary techniques used in titanium alloy welding. Such comparative analysis is essential for selecting suitable auxiliary methods for specific welding applications.

4.4. Comparative analysis of CMT, MIG, and laser-MIG hybrid processes

4.4.1. Overview and scope of comparison

In this section, we evaluate Cold Metal Transfer (CMT), Metal Inert Gas (MIG), and Laser-MIG hybrid welding with respect to titanium and its alloys (primarily Ti-6Al-4V, except as noted). The intention is to

provide an amalgamated evidence-based comparison along the following performance dimensions: thermal input and thermal input control; weld pool behaviour and metal transfer; microstructure and phase changes; mechanical properties (tensile, yield, ductility, toughness); residual stress and distortion; defect creation (porosity, cracking, lack of fusion, intermetallics); corrosion and surface integrity; production rate and energy/cost adjustment; and suitability for applications in industry with respect to joining and additive manufacturing (WAAM). Whenever applicable, we identify quantitative comparisons; where no numerical comparison exists, analogies are curated as either from controlled studies or bigger reviews [28,35,219]. Fig. 18 shows the key differences among CMT, MIG, and Laser-MIG hybrid welding processes. CMT operates with low heat input and stable short-circuit transfer, whereas MIG exhibits higher heat input with globular or spray transfer. The Laser-MIG hybrid process provides localized energy input and improved transfer stability. These variations influence microstructure, where CMT produces coarse $\alpha+\beta$ lamellae, MIG forms Widmanstätten structures, and Laser-MIG hybrid results in refined martensitic phases. Consequently, MIG generates higher residual stress and distortion, while CMT and Laser-MIG hybrid significantly reduce these effects, making them more suitable for titanium alloy applications.

4.4.2. Process microstructure property relationships

The welding behavior of titanium alloys is strongly governed by the interaction between process parameters, thermal cycles, and resulting microstructural transformations. In particular, heat input and cooling rate play critical roles in determining the formation of α , β , and martensitic α' phases in the weld region [220,221].

Low heat input processes such as CMT promote relatively high cooling rates during solidification. This thermal condition favours the formation of fine acicular α' martensite within the prior β grains of the fusion zone. The refined martensitic microstructure typically increases local hardness and tensile strength, while maintaining acceptable ductility when appropriate post-weld heat treatment is applied [222].

Conversely, welding processes involving higher heat input, such as conventional MIG welding, allow greater grain growth within the fusion zone and heat-affected zone. The slower cooling rate promotes the formation of coarser α -lamellae and larger prior- β grains. Such microstructural coarsening can reduce mechanical performance by decreasing yield strength and increasing susceptibility to crack initiation under cyclic loading [223].

Laser-MIG hybrid welding produces a unique thermal environment characterized by concentrated laser energy combined with arc heating. The resulting steep thermal gradients and rapid solidification promote fine microstructural features while maintaining sufficient filler metal deposition for joint stability. Consequently, hybrid welded joints frequently exhibit improved combinations of tensile strength and ductility compared with conventional arc welding processes [224].

The synthesis of these studies demonstrates that optimizing process parameters such as wire feed rate, welding speed, arc current, and laser defocusing distance is essential for controlling the thermal cycle and achieving desirable microstructural characteristics in titanium alloy

Table 10

Comparative analysis of auxiliary control techniques used in titanium alloy welding.

Auxiliary Technique	Applicable Welding Process	Mechanism	Control Effect	Limitations
Ultrasonic vibration	CMT + P, MIG	Acoustic cavitation and molten pool stirring	Grain refinement, suppression of columnar dendrites, improved tensile strength	Equipment complexity and process instability at high vibration intensity
Magnetic field assistance	Laser-MIG hybrid, MIG	Lorentz force modifies arc plasma and droplet trajectory	Improved arc stability, enhanced droplet detachment efficiency, reduced defects	Excessive magnetic field can destabilize arc
Shielding gas modification	MIG, CMT	Changes plasma conductivity and arc characteristics	Improved arc stability and droplet transfer	Excess oxygen causes weld embrittlement
Laser defocusing control	Laser-MIG hybrid	Adjusts energy density and keyhole stability	Reduced porosity and improved weld morphology	Excess defocusing reduces penetration
Groove geometry optimization	Laser-MIG hybrid	Alters plasma confinement and heat transfer	Increased penetration depth and weld stability	Complex groove preparation required

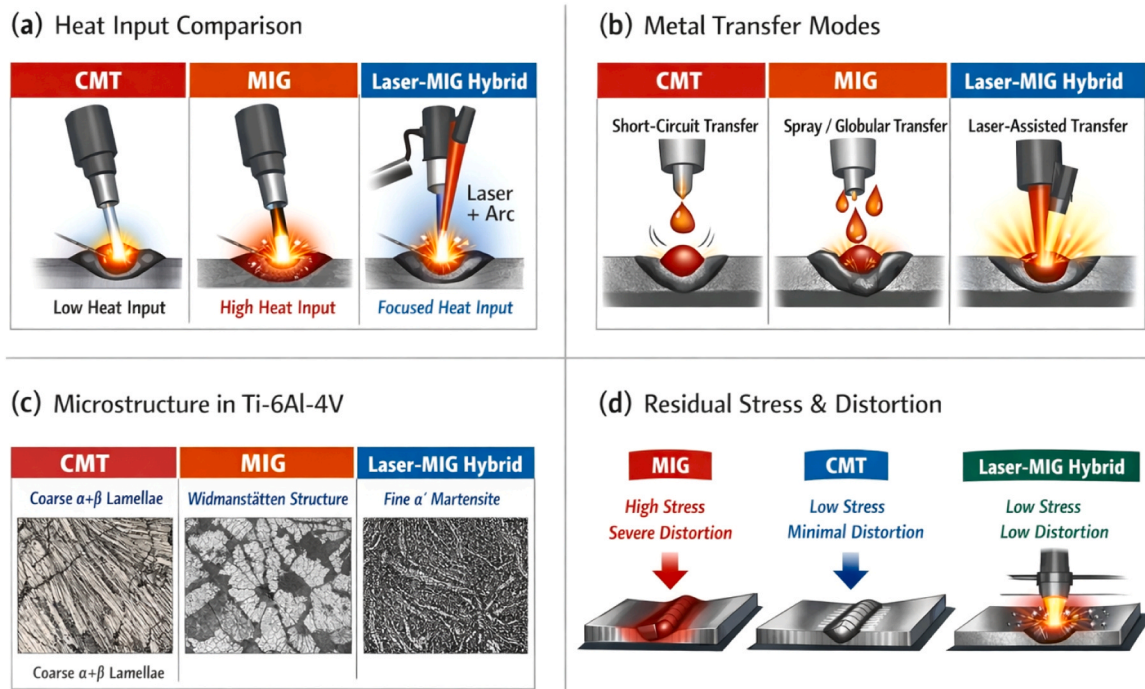


Fig. 18. Comparative schematic of CMT, MIG, and Laser-MIG hybrid welding for Ti-6Al-4V showing differences in (a) heat input, (b) metal transfer modes, (c) microstructure, and (d) residual stress and distortion.

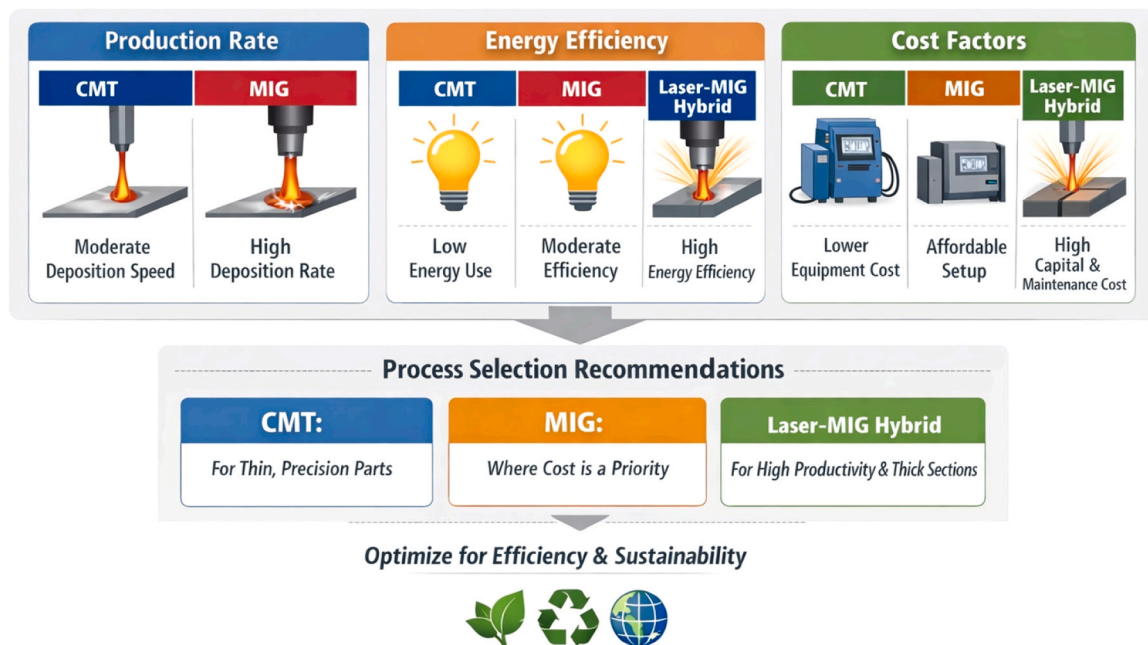


Fig. 19. Comparative analysis of production rate, energy efficiency, and cost for CMT, MIG, and Laser-MIG hybrid welding processes. MIG and Laser-MIG hybrid exhibit higher deposition rates, while CMT offers lower energy input. Laser-MIG hybrid demonstrates superior energy efficiency but incurs the highest capital cost, whereas MIG remains the most cost-effective option. Process selection guidelines for industrial applications are also highlighted.

welds.

4.4.3. Technical bottlenecks in titanium alloy welding

Despite significant progress in advanced welding technologies, several technical challenges remain in achieving consistently defect-free titanium welds.

One of the primary issues is porosity formation during welding. Titanium exhibits strong chemical affinity for oxygen, nitrogen, and

hydrogen at elevated temperatures. Improper shielding or unstable keyhole behavior during laser-based welding processes can lead to gas entrapment within the molten pool, resulting in porosity that degrades mechanical properties.

Residual stress generation is another critical concern. The steep thermal gradients associated with fusion welding processes can induce significant tensile residual stresses in the weld region. These stresses may lead to distortion, reduced fatigue resistance, and potential

cracking in structural components. Although low-heat-input processes such as CMT reduce residual stress compared with conventional arc welding, complete elimination of thermal stresses remains challenging.

The formation of brittle intermetallic compounds also presents difficulties in dissimilar metal welding involving titanium. When titanium is joined with materials such as steel or aluminum, intermetallic phases such as TiFe or TiAl₃ can form at the interface. These phases are typically brittle and may significantly reduce joint toughness.

Another challenge involves achieving uniform microstructure throughout the weld region. Rapid solidification can produce heterogeneous microstructures containing mixtures of α' martensite and retained β phase. Such microstructural gradients may lead to variations in hardness and mechanical performance across the weldment.

Addressing these technical bottlenecks requires a deeper understanding of weld pool dynamics, improved process monitoring techniques, and optimization of process parameters to control heat input and solidification behavior.

4.4.4. Heat input, thermal control, and process efficiency

Heat input is the primary factor that governs microstructural transformations, residual stress generation, distortion, and the likelihood of defects such as intermetallics and contamination in titanium welding. CMT is characterized by exceptionally low, well controlled heat input relative to conventional MIG due to its sequence of short-circuiting, wire retraction and controlled arc energy transfer; this reduces peak temperature and thermal gradient around the fusion zone. Multiple recent reviews and experimental studies report that CMT produces lower overall heat input and smaller HAZs compared to standard MIG, enabling welding of thinner sections with minimal distortion [225,226].

By contrast, conventional MIG (globular or spray transfer modes) inherently introduces higher heat input unless operated with pulsed modes specifically optimized for heat control; pulsed-MIG reduces average power but usually not to the level of CMT. MIG's higher heat input tends to enlarge the HAZ and promote coarser microstructures in the fusion zone when compared at similar travel speeds [47,227].

Laser-MIG hybrid welding couples a high-power laser (concentrated energy, deep penetration) with an arc (filler addition, process stability). When optimized, the hybrid approach can achieve **higher specific welding speeds and deeper penetration with a lower net heat input per unit length** than MIG, approaching or exceeding the productivity of MIG while maintaining a relatively narrow HAZ comparable to or smaller than MIG alone [228–230]. In many studies across alloys (including titanium and aluminum analogs), Laser-MIG hybrid demonstrates superior energy efficiency and higher welding rates with controlled thermal input, provided laser and arc are well aligned and process parameters tuned. However, achieving this requires careful control of laser power, focal position, arc-laser distance, and shielding all more demanding than CMT or MIG in terms of setup [231–234].

For minimal heat input and low distortion in thin sections, CMT is preferable; for high productivity with controlled HAZ and deep penetration, Laser-MIG hybrid can outperform MIG and approach CMT's advantages when parameters are optimized. MIG (non-pulsed) generally has the highest heat input and therefore the largest HAZ and distortion.

4.4.5. Metal transfer and weld pool dynamics

Metal transfer behavior strongly influences spatter, weld bead geometry, pore formation and oxide entrainment all critical for titanium's reactivity and contamination sensitivity [235].

CMT uses a distinctive mechanically assisted metal transfer (wire retraction during short-circuit and a quasi-cold arc) that yields predictable droplet detachment at low energy. This yields minimal spatter and a stable droplet size distribution beneficial for titanium where surface contamination via atmospheric entrainment must be strictly controlled. Several experimental studies show smoother bead surfaces and lower spatter with CMT relative to MIG for the same wire and shielding gas combinations [21,34,236,237].

MIG in globular mode generates larger droplets and more spatter than either pulsed-MIG or CMT, increasing risk of contamination and surface defects. Pulsed MIG improves transfer stability and reduces spatter, but still often falls short of CMT's precision at low currents. For titanium, especially thinner sheets or where weld surface finish and contamination control are paramount (e.g., biomedical components), pulsed transfer can be acceptable but requires precise shielding and process control [233].

Laser-MIG hybrid creates complex interaction phenomena: the laser produces a keyhole and deep penetration, the arc supports filler metal deposition and stabilizes the weld pool, and interactions can lead to improved wetting and reduced porosity when tuned correctly. However, hybrid interactions can also create unstable behaviour if alignment, shielding, or filler feed is poor leading to spatter or entrainment of oxides. Studies demonstrate that hybrid modes often refine microstructure in the fusion zone due to higher cooling rates from the concentrated energy density, but the metal transfer and pool dynamics are more parameter sensitive than either standalone process [192,232,238,239].

CMT provides the most controlled, low-spatter metal transfer; pulsed MIG improves but does not always match CMT; hybrid welding offers excellent filler integration and penetration but is most sensitive to setup and parameter variation [240].

4.4.6. Microstructural evolution and phase transformations

Titanium alloys, especially Ti-6Al-4V, exhibit microstructural sensitivity to peak temperature and cooling rates that control α/β phase fractions, grain size, and martensitic transformations (α' formation) all of which determine mechanical properties [241,242].

- **CMT's** low heat input and relatively higher cooling rates favor formation of fine α or α' martensitic structures in the fusion zone, with narrow HAZs. The finer microstructures typically increase hardness locally but can be tempered by post-weld heat treatments. Reports indicate CMT welds often show finer columnar or acicular α structures and smaller prior- β grain sizes compared to MIG welds run at higher heat input [35,179,243,244].
- **MIG** (higher heat): larger prior- β grains, broader HAZ, and more pronounced coarsening of α lamellae. When pulsed MIG is used with low average heat input, microstructural coarsening is reduced and properties improve, but still can lag behind CMT for minimising HAZ width [245].
- **Laser-MIG hybrid:** the concentrated laser input produces steep thermal gradients and rapid solidification, often leading to a narrow fusion zone with refined microstructure (fine columnar grains, reduced dendrite sizes) and suppressed HAZ coarsening. Hybrid processes can therefore achieve a favourable strength ductility balance in many studies though the exact microstructure depends strongly on process synergy (laser power vs arc energy) [224,246]. Recent reviews on laser welding of titanium report that laser and hybrid methods produce narrower HAZ and more desirable microstructures for fatigue and corrosion resistance when compared to traditional arc welding [247,248].

It is important to note that most studies focus on Ti-6Al-4V; however, other titanium alloy systems exhibit different responses due to compositional variations. For instance, CP-Ti shows greater sensitivity to interstitial impurities, whereas β -rich alloys demonstrate enhanced hardenability and retained β -phase stability. Similarly, alloys such as Ti-Al-Zr-Fe exhibit improved weldability due to modified solidification behavior and reduced cracking susceptibility. These variations highlight the need to interpret welding outcomes in the context of alloy chemistry, rather than generalizing results based solely on Ti-6Al-4V [27,249].

- **Quantitative comparisons:** Where comparable process windows have been reported, CMT and Laser-MIG hybrid welds often show smaller HAZ widths (by tens to hundreds of micrometres) and finer α lamellar spacing than MIG welds under higher heat input; however,

absolute numbers depend on plate thickness, travel speed, and shielding conditions reported in the specific studies cited [179,240,250].

4.4.7. Mechanical properties: strength, ductility, toughness, and hardness

Quantitatively, welded Ti-6Al-4V joints produced by CMT and Laser-MIG hybrid processes typically achieve UTS values in the range of ~900–1050 MPa (Table 7), approaching or exceeding 90–100 % of base metal strength. In contrast, conventional MIG welds often show reduced strength (~270–410 MPa for CP-Ti systems) due to higher heat input and coarser microstructures. Similarly, elongation values in hybrid welds (>10–15 %) are generally higher than those in conventional arc welds, indicating improved ductility [165,251]. Mechanical outcomes depend on microstructure and defect content. The literature shows consistent trends across comparative studies:

- **Tensile strength and yield:** CMT and Laser-MIG hybrid welds typically retain a higher fraction of the base metal (BM) strength compared to conventional MIG welds when matched by travel speed, largely due to reduced HAZ coarsening and fewer defects. Several comparative investigations (including those that benchmark CMT against MIG and laser hybrids) report that CMT and Laser-MIG bonds deliver equal or superior ultimate tensile strength (UTS) and yield strength than MIG, especially when process parameters are optimized [179].
- **Ductility and toughness:** Ductility is more sensitive to defect population (porosity, oxides) and brittle intermetallic presence. Laser-MIG hybrid, with proper shielding and parameter choice, often achieves superior ductility compared to MIG due to a cleaner fusion zone and narrower HAZ; CMT can produce comparable ductility although locally increased hardness from fine α' may reduce elongation slightly if not tempered. Overall, well-executed CMT and hybrid processes generally equal or exceed pulsed-MIG in ductility metrics [47,252,253].
- **Hardness distribution:** CMT welds commonly show higher hardness peaks in the fusion zone due to rapid cooling and martensitic α' formation; Laser-MIG hybrid can show similar or lower hardness depending on heat partitioning. MIG welds show more graded hardness profiles corresponding to larger HAZ and tempering effects. Hardness must be correlated with toughness; elevated hardness alone does not guarantee better structural performance [47].
- **Fatigue:** Fatigue performance is dominated by surface integrity and subsurface defects; hybrid processes that minimize porosity and produce smoother beads often show improved fatigue life. Some WAAM/CMT additive studies indicate promising fatigue for CMT-based builds after suitable post-processing, but fatigue comparisons for fusion welding are highly dependent on surface finishing and residual stresses [254–258].

When carefully controlled, CMT and Laser-MIG hybrid welding produce joints with mechanical properties that meet or exceed modern pulsed-MIG. The deciding factors are defect control and post-weld treatments (e.g., stress relief, annealing) that modify hardness and toughness profiles.

4.4.8. Long-term performance and fatigue relevance for aerospace applications

Beyond static tensile strength and hardness, the long-term structural reliability of titanium welds in aerospace service is governed by fatigue crack initiation resistance, crack propagation behavior, residual stress state, and environmental stability. In general, fatigue performance is strongly influenced by weld toe geometry, porosity, surface oxidation, microstructural heterogeneity, and subsurface defects. From this perspective, CMT-based welding offers important advantages because its low heat input and controlled droplet transfer produce narrower heat-affected zones, reduced distortion, and lower residual stress

accumulation, all of which are beneficial for delaying fatigue crack initiation [259]. In addition, the comparatively smoother bead profile and low-spatter transfer characteristics of CMT can reduce local stress concentration sites. In contrast, conventional MIG welding generally introduces higher heat input, wider fusion and heat-affected zones, and coarser transformed microstructures, which may reduce fatigue resistance, particularly when porosity, oxide contamination, or weld toe irregularities are present. Laser-MIG hybrid welding appears especially promising for aerospace titanium structures because the combined action of the laser and arc can produce deeper penetration, smoother bead morphology, lower porosity, and a more favorable strength-ductility combination than conventional MIG alone [27,260]. These features are expected to improve service longevity by suppressing early crack nucleation and slowing crack growth. However, the fatigue response of hybrid joints remains sensitive to keyhole stability, process alignment, and shielding quality. Therefore, although available literature indicates that CMT and Laser-MIG hybrid processes offer superior fatigue potential compared with conventional MIG, systematic long-term fatigue, corrosion-fatigue, and thermomechanical durability studies under aerospace-relevant service conditions are still limited and remain necessary for qualification of welded titanium components [251,261].

4.4.9. Residual stresses and distortion

Residual stress formation is governed by the thermal cycle and mechanical constraints. Lower heat input and smaller HAZs reduce the magnitude of thermally induced residual stresses and the resultant distortion [262].

- **CMT:** Several comparative studies report that CMT introduces lower peak tensile residual stresses and reduced distortion versus MIG due to its lower heat input and slower heat accumulation. For titanium components requiring tight tolerances, CMT's reduced distortion is a clear advantage. However, reported residual stress magnitudes can vary instrument-to-instrument and study-to-study; trends are more reliable than absolute numbers [263,264].
- **MIG:** Higher residual stresses and greater distortion are typical for MIG welding on account of higher heat input. Pulsed MIG reduces but does not eliminate this issue. Residual stress measurement methods (XRD, hole-drilling, neutron diffraction) and sampling locations strongly influence reported magnitudes so comparative assessments should account for measurement method variability [265–270].
- **Laser-MIG hybrid:** The hybrid process usually results in narrower, more localized residual stress fields due to concentrated laser heating and rapid cooling; with proper parameter choice, peak residual stresses can be comparable to or lower than MIG. However, localized high thermal gradients near the keyhole can create steep stress gradients requiring careful welding paths and fixturing to manage distortion [271–274].

Measurement caveat: As emphasized in recent literature, residual stress results are instrument-sensitive and can vary significantly by measurement technique and specimen preparation; hence comparative statements should rely on studies that use the same measurement protocol [275–277].

4.4.10. Defect formation, contamination risk, and shielding requirements

Titanium's high affinity for oxygen, nitrogen, and hydrogen at elevated temperature makes shielding quality decisive.

- **CMT** typically produces low spatter and smaller, more stable pools, reducing the risk of oxide entrainment. However, all processes require rigorous inert shielding (argon $\pm$ helium mixes) and proper trailing nozzles or backing gas to avoid embrittling pickup. CMT benefits from lower overall weld metal exposure time at high temperatures [34,278,279].

- **MIG** exposes more molten metal area and time at elevated temperature (unless carefully pulsed), increasing the probability of contamination if shielding is inadequate. Spatter from globular transfer can deposit oxides. Pulsed MIG with optimized shielding and filler chemistry can mitigate this risk.
- **Laser-MIG hybrid** can lower contamination by reducing the time and volume of molten metal, but the keyhole can be a site for turbulence and gas entrapment if shielding/gas flow is poorly designed. Hybrid welding's higher travel speeds shorten high-temperature exposure and thereby reduce contamination risk in many studies again, provided shielding is engineered appropriately [280–283].

4.4.11. Intermetallics and dissimilar joins

For dissimilar welding (e.g., Ti to Al or steel), hybrid processes that minimize dilution and control thermal cycles can better limit intermetallic growth; but specialized filler strategies and joint design remain necessary [35,64,284].

4.4.12. Corrosion resistance and surface integrity

Corrosion performance of welded titanium alloys is primarily determined by microstructure, residual stress, and contamination. Narrow HAZs, absence of oxygen/nitrogen pickup, and minimal porosity favor corrosion resistance [285–288].

- **CMT and Laser-MIG hybrid** welds typically show corrosion behaviour closer to base metal than standard MIG when shielding and heat input are controlled, due to reduced contamination and narrower HAZs. Studies of laser and hybrid welds on Ti alloys confirm improved surface integrity and reduced susceptibility to localized corrosion when processing is optimized [27].

Takeaway: Clean, low-heat processes (CMT or hybrid) favours corrosion performance but shielding discipline is indispensable across all processes.

4.4.13. Production rate, energy use, and cost analysis

- **Production rate:** MIG (especially high-power spray MIG) and Laser-MIG hybrid typically offer higher deposition rates or travel speeds than CMT in single-pass joining of equivalent thicknesses. Laser-MIG is particularly advantageous where deep penetration and single-pass welding of thicker plates are required. CMT's deposition rate is often lower, but its minimal rework requirements and low distortion may offset slower deposition in production scenarios that value dimensional accuracy [289–292].
- **Energy efficiency:** Laser-MIG hybrid often shows high energy efficiency per unit weld length due to the laser's concentrated energy and resultant high travel speeds. CMT's low absolute energy input can make it energetically efficient for thin sections and delicate assemblies. Comparative life-cycle cost (LCC) depends strongly on part geometry, throughput requirements, and post-weld treatments [293].
- **Equipment and operating costs:** CMT equipment (specialized power sources and wire feed systems) can be more expensive than basic MIG setups but generally cheaper than a laser-hybrid cell which requires high-power lasers, beam delivery and safety infrastructure. Laser-MIG hybrid systems command the highest capital expenditure and require higher operator skill and maintenance. For high value or high-precision aerospace/medical components the premium may be justified [240].

In addition to performance advantages, economic and scalability considerations play a decisive role in process selection. Laser-MIG hybrid and LBW systems involve significantly higher capital and maintenance costs compared to conventional MIG or CMT setups, limiting their adoption to high-value applications such as aerospace and

biomedical industries. EBW, while offering exceptional weld quality, is restricted by vacuum chamber size and operational constraints, making it less suitable for large structural components. Moreover, the requirement for precise parameter control and skilled operators further increases operational complexity. These limitations suggest that, despite superior metallurgical performance, advanced welding processes must be evaluated in terms of cost-efficiency, scalability, and industrial feasibility.

Choose CMT for low-distortion, high-quality, thin components (moderate throughput). Choose Laser-MIG hybrid for high speed, deep penetration, and high productivity despite higher capital cost. Use MIG where cost constraints dominate and component tolerances/requirements are less stringent. From a sustainability standpoint, processes that combine low heat input with high material utilization, such as CMT and optimized Laser-MIG hybrid welding, offer significant potential for reducing overall environmental impact and improving life-cycle efficiency of welded structures [240,294]. Fig. 18 and Table 11 highlights that MIG and Laser-MIG hybrid processes achieve higher deposition rates, while CMT prioritizes low heat input and precision. Laser-MIG hybrid offers superior energy efficiency but involves high capital cost. MIG is cost-effective, whereas CMT is suitable for high-quality, low-distortion applications.

4.4.14. Economic and Supply Chain Considerations in Titanium Alloy Welding

The widespread industrial adoption of titanium alloys is significantly influenced by economic factors, including raw material cost, processing complexity, and supply chain constraints. Titanium production is inherently expensive due to the energy-intensive Kroll process, which involves multiple stages of chlorination, reduction, and purification. Consequently, titanium alloys such as Ti-6Al-4V are substantially more costly than conventional structural materials like steels or aluminum alloys [295,296].

In addition to production costs, the availability of titanium is dependent on mineral resources such as rutile and ilmenite, which are geographically concentrated, leading to potential supply chain vulnerabilities. Fluctuations in global demand, geopolitical factors, and export restrictions can further influence material availability and pricing, particularly in aerospace-grade titanium. These economic constraints directly affect the selection of welding processes. Advanced techniques such as Cold Metal Transfer (CMT) and Laser-MIG hybrid welding, while offering superior microstructural control and reduced defect formation, require higher capital investment in specialized equipment and process control systems. In contrast, conventional MIG welding remains more economically viable for large-scale applications despite its limitations in weld quality [297].

Table 11
Comparative analysis of production rate, energy use, and cost.

Parameter	CMT	MIG	Laser-MIG Hybrid
Production rate	Moderate	High	Very high
Travel speed	Low-moderate	High	Very high
Heat input	Very low	High	Moderate (localized)
Energy efficiency	High (thin sections)	Moderate	Very high
Distortion/ rework	Minimal	High	Very low
Equipment cost	Moderate	Low	Very high
Operating cost	Moderate	Low	High
Maintenance complexity	Moderate	Low	High
Best application	Thin, precision parts	General industrial use	Thick sections, high productivity
Industrial suitability	Aerospace, biomedical (precision)	Cost-sensitive manufacturing	High-value, high-performance sectors

From an industrial perspective, the trade-off between cost and performance is a key consideration. High-value sectors such as aerospace and biomedical engineering justify the use of advanced welding technologies due to stringent performance requirements. Moreover, emerging approaches such as CMT-based Wire Arc Additive Manufacturing (WAAM) offer improved material utilization efficiency, reduced waste, and lower buy-to-fly ratios, thereby enhancing the overall economic feasibility of titanium structures [298].

Therefore, integrating economic and supply chain considerations with process optimization is essential for enabling the broader adoption of titanium alloy welding technologies in industrial applications.

4.4.15. Industrial scalability and economic considerations

The industrial adoption of welding processes for titanium alloys is strongly governed not only by metallurgical performance but also by scalability, cost efficiency, and production feasibility. While laboratory-scale studies emphasize microstructural refinement and mechanical performance, industrial implementation requires balancing these attributes with throughput, automation capability, and economic viability. From a scalability perspective, MIG welding remains the most widely adopted process in industrial environments due to its high deposition rate, relatively low equipment cost, and ease of integration with automated and robotic systems. Its capability for continuous operation and adaptability to large structures makes it suitable for mass production, particularly in automotive and structural applications where ultra-high precision is not mandatory [299]. Fig. 20 shows that MIG is suitable for low-cost, high-volume production, while CMT is preferred for precision and low-distortion applications. Laser-MIG hybrid offers high productivity and superior performance but at higher cost. Overall, process selection depends on balancing cost, scalability, and weld quality requirements.

In contrast, Cold Metal Transfer (CMT), although derived from MIG, prioritizes process stability and low heat input over deposition rate. This makes CMT particularly advantageous for thin sections, dissimilar material joining, and applications requiring minimal distortion, such as aerospace panels and biomedical components. However, its relatively lower productivity compared to conventional MIG may limit its direct use in high-throughput manufacturing unless combined with additive manufacturing approaches such as WAAM, where its controlled heat input becomes a significant advantage [34,300,301].

Laser-MIG hybrid welding (LAMIG) represents a technologically advanced solution that bridges the gap between productivity and weld quality. The combined action of a high-energy laser and arc source enables deep penetration, high welding speeds, and reduced heat-affected zones, making it highly suitable for medium-to-thick sections in aerospace and high-performance engineering structures. Nevertheless, the process involves high capital investment, complex parameter control, and stringent alignment requirements, which can limit its widespread adoption to high-value manufacturing sectors [219].

From an economic standpoint, the selection of welding process can be summarized as follows:

- > Low-cost, high-volume production: MIG welding
- > High-precision, low-distortion applications: CMT
- > High-performance, high-value industrial applications: Laser-MIG hybrid

Therefore, while advanced processes such as CMT and LAMIG offer superior metallurgical and mechanical performance, their industrial scalability depends on application-specific trade-offs between cost, productivity, and quality requirements. Future developments in process automation, hybridization, and digital control are expected to further enhance the scalability and industrial feasibility of these advanced welding technologies.

4.4.16. Suitability for additive manufacturing and repair (WAAM, DED)

Recent work has actively combined CMT with directed energy and WAAM approaches for Ti-6Al-4V deposition. CMT-WAAM offers lower heat input per layer, reduced residual stress accumulation and finer as-deposited microstructures. Laser-MIG hybrid directed energy deposition also shows promise for high deposition rates with improved geometrical accuracy but requires sophisticated process control to avoid keyholing or excessive dilution. MIG-based WAAM without controlled metal transfer often suffers from higher heat accumulation and thus greater distortion and residual stress [87,302–304].

CMT-based WAAM is attractive for titanium WAAM where dimensional control and reduced heat accumulation matter; laser-hybrid DED offers higher deposition rates but increases process complexity [304, 305].

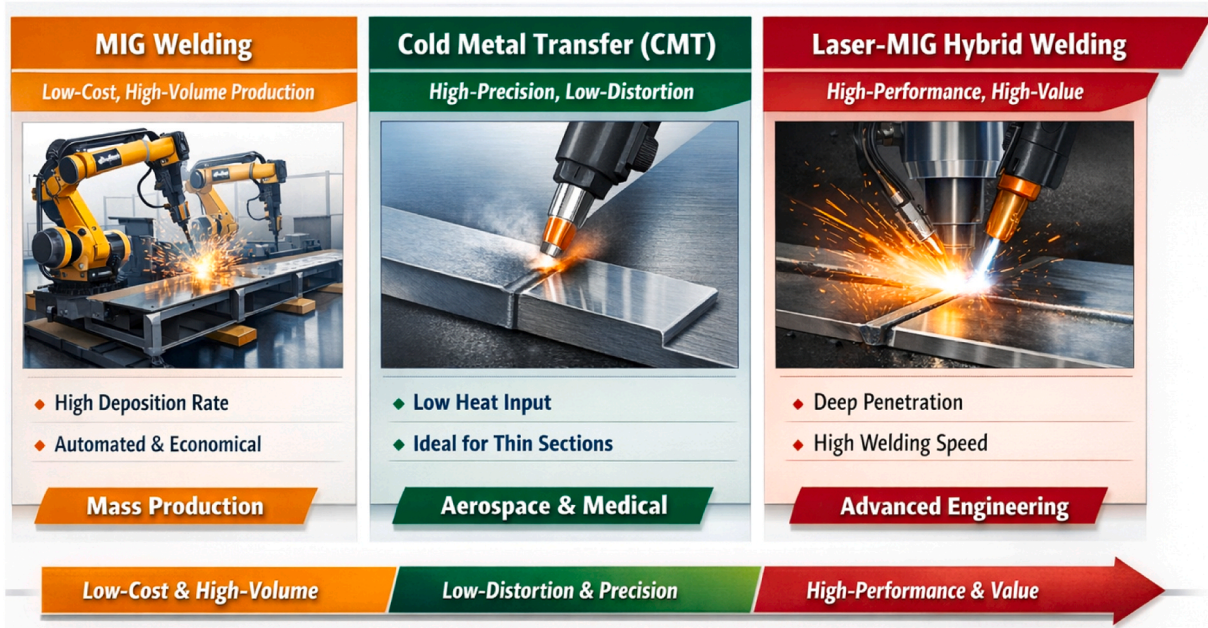


Fig. 20. Decision framework for selecting CMT, MIG, and Laser-MIG hybrid welding based on cost, productivity, and scalability for titanium alloys.

4.4.17. Environmental, safety and practical considerations

- **Shielding gas consumption:** CMT and pulsed processes can be optimized for lower gas usage; hybrid setups may require more complex shielding geometries and gas management, especially when using helium-argon blends to stabilize arcs and lasers [306,307].
- **Operator safety and infrastructure:** Laser-hybrid requires stringent laser safety protocols, beam enclosures and higher capital expenditure; CMT and MIG are comparatively straightforward but require good ventilation and handling measures to manage fumes and spatter.
- **Environmental and Sustainability Considerations in Titanium Welding Processes:** In recent years, sustainability has emerged as a key consideration in advanced manufacturing processes, including welding of titanium alloys. The environmental impact of welding processes is governed by factors such as energy consumption, shielding gas usage, material efficiency, and lifecycle performance of welded structures. Among the processes reviewed, Cold Metal Transfer (CMT) demonstrates a significant advantage in terms of energy efficiency due to its low heat input mechanism. The controlled short-circuiting and wire retraction cycle reduce total energy consumption per unit weld length, thereby lowering the carbon footprint of the process [22,308]. Additionally, reduced spatter formation in CMT improves material utilization and minimizes post-weld cleaning operations, further contributing to sustainability. In contrast, conventional MIG welding typically involves higher heat input and increased fume generation, leading to greater energy consumption and environmental burden. Although pulsed MIG reduces average power input, it still exhibits comparatively higher thermal loads and shielding gas consumption. The generation of welding fumes and spatter in MIG processes also necessitates additional ventilation and waste handling systems. Laser-MIG hybrid welding offers a balanced approach, combining high energy density laser input with arc welding flexibility [34,309]. This process can achieve higher welding speeds and deeper penetration, thereby reducing energy consumption per unit length of weld. However, the overall environmental impact depends on the efficiency of the laser system and the complexity of shielding gas requirements, particularly when helium-based mixtures are employed. From a lifecycle assessment (LCA) perspective, processes with lower heat input and minimal defect formation, such as CMT and optimized hybrid welding, are advantageous as they reduce rework, material waste, and energy-intensive post-processing steps. Moreover, improved weld quality and longer service life of components contribute to overall sustainability in aerospace and biomedical applications [310, 311].

Future research should focus on developing environmentally optimized welding strategies, including low-energy arc modes, alternative shielding gases, and integration of real-time monitoring to minimize defects and resource consumption.

4.4.18. Practical guidance and recommended parameter strategies

From the literature synthesis, recommended practical strategies for each process when welding titanium alloys:

- **CMT:** Use when minimal distortion and clean surface finish are priorities (thin sections, complex geometry). Optimize wire feed speed and short-circuit parameters; ensure trailing/backing gas coverage. Consider post-weld stress relief for high hardness in fusion zone [236,278].
- **Pulsed MIG:** Employ where higher deposition is needed but CMT is unavailable; use pulsed modes to lower average heat and minimize contamination; strict shielding and cleaning of joint surfaces are essential.

- **Laser-MIG hybrid:** Best for deep penetration, high travel speeds, and thicker sections in production. Control laser/arc alignment and focal position; hybrid-specific shielding and synchronized wire feed enable superior bead quality and minimal porosity. For dissimilar joints or structures demanding minimal HAZ, hybrid is promising but requires precise fixturing [231].

5. Knowledge gaps, limitations, and research needs

Despite advances, several gaps impede direct, universally applicable recommendations:

1. **Standardized comparative datasets:** Many studies use different base materials, thicknesses, filler chemistries, and measurement protocols (especially for residual stress). There is a need for multi-laboratory standard test matrices comparing CMT, MIG (pulsed), and Laser-MIG on identical materials and geometry using the same measurement standards [312–314].
2. **Long-term performance (fatigue/corrosion) under service conditions:** Most comparisons focus on static properties; systematic fatigue, fretting, and corrosive environment studies over long durations are fewer and necessary for aerospace/biomedical qualification [315–317].
3. **Process modeling and in-situ control:** Hybrid and CMT processes would benefit from predictive models that couple fluid flow, phase transformation kinetics, and real-time control (AI/ML). Early studies are promising, but broader validation is required [318,319].
4. **Dissimilar metal joining:** Laser-MIG hybrid shows promise in controlling intermetallic formation, but optimized filler strategies and interlayer designs are needed for robust dissimilar joints (e.g., Ti-Al, Ti-steel) [190,320,321].

5.1. Final comparative summary (practical decision matrix)

- **Quality (microstructure & contamination):** CMT $\approx$ Laser-MIG hybrid $>$ pulsed-MIG $>$ globular MIG.
- **Heat input and distortion:** CMT $<$ Laser-MIG hybrid $<$ pulsed-MIG $<$ globular MIG.
- **Productivity (speed and penetration):** Laser-MIG hybrid $\geq$ MIG $>$ CMT (context dependent).
- **Equipment cost and complexity:** Laser-MIG hybrid $>$ CMT $>$ MIG.
- **Best use cases:** CMT for thin, precision, low-distortion components and CMT-WAAM; Laser-MIG hybrid for high throughput, deep single-pass joints; MIG for cost-sensitive, less critical joints.

6. Conclusion

- Cold Metal Transfer (CMT) and its pulse/ultrasonic-assisted variants demonstrate remarkable advantages over conventional welding techniques by offering low heat input, spatter-free metal transfer, and refined weld microstructures, enabling reliable joining of titanium alloys.
- Ultrasonic vibration during CMT+P welding effectively inhibits columnar grain growth and refines α' martensite, leading to improved ultimate tensile strength (UTS) and mixed ductile brittle fracture modes in Ti-6Al-4V joints.
- Optimizing shielding gas composition (Ar + O₂/CO₂) in Pulsed MIG welding enhances arc stability and droplet transfer behavior, improving tensile strength and hardness. However, excessive oxidizing gas introduces brittleness and must be precisely controlled.
- LAMIG welding provides a balance between laser precision and arc adaptability. Optimized defocusing distance (+6 to +12 mm) and magnetic field control (≈ 24 mT) significantly reduce porosity ($>80\%$), improve plasma stability, and ensure defect-free welds.

- Adjusting groove geometry and process parameters allows control of plasma temperature, penetration depth, and droplet frequency, resulting in superior weld morphology and consistent bead formation in medium to thick titanium plates.
- Comparative analysis reveals that hybrid techniques generate finer α -lamellae and reduced β -phase volume, correlating with higher strength, ductility, and corrosion resistance compared with conventional GMAW or TIG processes.
- Titanium alloys demonstrate clear performance superiority over conventional materials such as steels, aluminum, and nickel-based alloys in terms of specific strength, corrosion resistance, and biocompatibility, justifying their increasing adoption in critical industries.
- The superior arc control, minimal porosity, and excellent strength-ductility balance of CMT and LAMIG make these techniques particularly suited for aerospace, defense, biomedical, and energy applications where reliability and precision are critical.
- The evolution of the heat-affected zone (HAZ) is strongly dependent on welding heat input, where CMT produces the narrowest HAZ with refined microstructure, MIG results in wider HAZ and grain coarsening, and Laser-MIG hybrid achieves an optimal balance, directly influencing weld strength, ductility, and long-term performance.
- A direct comparison indicates that ultrasonic-assisted CMT + P welding achieves significantly higher tensile strength (≈ 1000 MPa) than pulsed MIG welding (≈ 300 – 400 MPa), primarily due to refined microstructure and reduced heat input, while MIG welds show reduced ductility and increased brittleness with higher oxidizing gas content.
- The compatibility of dissimilar materials, particularly in titanium-based systems, is strongly influenced by intermetallic formation, thermal expansion mismatch, and differences in mechanical properties. Advanced low-heat-input processes such as CMT and Laser-MIG hybrid welding, combined with appropriate filler and inter-layer strategies, are essential for achieving reliable dissimilar joints.
- In addition to summarizing existing research, this review establishes a comparative framework linking welding process parameters, microstructural evolution, and mechanical performance in titanium alloy joints. By identifying common trends, process-specific advantages, and current technological limitations, the study provides a foundation for future development of advanced welding technologies for aerospace, biomedical, and energy applications.

7. Challenges and research gaps

Despite notable progress in the field, several scientific and technological challenges persist in achieving consistent, defect-free titanium welds, especially for critical aerospace and biomedical applications.

• Control of Porosity and Keyhole Instability:

Although defocusing distance optimization significantly reduces porosity, full elimination remains difficult due to rapid vaporization and collapse of the keyhole in LAMIG welding [322–324]. Advanced in-situ diagnostics and modeling are required to correlate melt pool dynamics with gas entrapment mechanisms [325,326].

• Mechanical and Fatigue Behavior:

Although static tensile and hardness data are widely reported, long-term fatigue performance data for CMT, MIG, and Laser-MIG hybrid welded titanium joints remain limited. This is particularly critical for aerospace structures, where cyclic loading, vibration, thermal fluctuation, and stress concentration at weld toes govern service life. CMT joints are expected to provide improved fatigue resistance due to reduced heat input, lower distortion, and finer microstructural features, whereas conventional MIG joints may be more vulnerable to fatigue crack initiation because of wider HAZ, coarser transformed structures, and higher tensile residual stresses. Laser-MIG hybrid joints show strong potential for improved

longevity because reduced porosity and smoother weld morphology can delay crack nucleation; however, their long-term performance is highly dependent on keyhole stability and parameter control. More systematic investigations correlating process conditions, defect population, residual stress, and cyclic deformation behavior are required for aerospace certification [327–329].

• Residual Stress and Distortion:

Even low-heat-input processes such as CMT produce residual stresses due to steep thermal gradients [330]. Real-time temperature mapping and finite element-based residual stress modeling are essential for quantifying and mitigating these effects, particularly in multi-pass and thick-section welds [331–335].

• Intermetallic Formation in Dissimilar Joints:

In dissimilar titanium-steel or titanium-aluminum systems, brittle intermetallics (TiFe, TiAl₃) can form at the interface, reducing joint toughness. Advanced filler design, graded transition layers, and precise thermal cycle control are promising strategies to suppress such formations [64,67,336–338]. Advanced filler design, graded transition layers, and precise thermal cycle control are promising strategies to suppress such formations while improving compatibility between dissimilar materials.

• Microstructural Homogeneity:

The weld region often exhibits a dual microstructure of α' martensite and retained β -phase, leading to property gradients across the weldment. Post-weld heat treatments (PWHT) and thermo-mechanical stabilization cycles require further optimization to ensure uniform grain morphology and phase balance [339–342].

• Droplet Transfer and Arc Stability Modeling:

Although waveform control in CMT has been well characterized, the coupled physics of current, wire motion, and metal transfer still lack complete theoretical models. Multiphysics simulations integrating magnetohydrodynamic and plasma effects could yield predictive capability for process control [278,343].

• Scaling to Industrial and Additive Systems:

Transferring laboratory-scale results to large aerospace components or wire arc additive manufacturing (WAAM) frameworks introduces new challenges in process repeatability, energy efficiency, and environmental contamination control [153,344–346].

• Mechanical and Fatigue Behavior:

Long-term fatigue and creep properties of CMT and LAMIG joints remain underexplored, especially under variable load and temperature conditions. Systematic investigations correlating microstructural refinement with cyclic deformation response are needed for certification of aerospace-grade welds [347,348].

• Corrosion and Biocompatibility Assessment:

For biomedical applications, understanding the corrosion mechanisms and ion leaching behavior of Ti-6Al-4V welds under physiological environments remains limited. Hybrid surface modifications and coating approaches could further enhance performance [82,349–352].

• Economic and Scalability Limitations of Advanced Welding Techniques:

While processes such as CMT, EBW, and LBW demonstrate superior control over heat input and microstructure, their industrial implementation is often constrained by high equipment cost, limited accessibility, and challenges in scaling to large or complex structures. Future research should focus on developing cost-effective hybrid systems, modular welding platforms, and process simplification strategies to enhance industrial adaptability [353,354].

• Titanium alloy dissimilar welding, especially for combinations such as Ti/Al and Ti/steel

A particularly important research gap remains in the area of titanium alloy dissimilar welding, especially for combinations such as Ti/Al and Ti/steel, where brittle intermetallic compounds, elemental segregation, and severe thermo-physical mismatch continue to limit

joint reliability. Current research increasingly focuses on minimizing reaction-layer thickness through low-heat-input processes, hybrid heat sources, interlayer engineering, and tailored filler design. For industrial adoption, future studies must establish stronger process-microstructure-property relationships for these dissimilar systems, with special attention to interfacial stability, residual stress evolution, and long-term mechanical performance [64,355].

• Influence of Alloy Composition and Impurities:

Most existing studies are heavily centered on Ti-6Al-4V, with limited exploration of how compositional variations affect weld behavior. Future research should focus on understanding the role of alloying elements and interstitial impurities in controlling phase transformations, residual stress evolution, and defect formation across different titanium alloy systems [356,357].

8. Future research directions

The continuous evolution of welding technologies for titanium and its alloys reflects the growing need for lightweight, high-strength, and corrosion-resistant joints in aerospace, biomedical, and defense applications. Future research must therefore focus on integrating process intelligence, advanced sensing, and hybrid energy sources to achieve precision and reproducibility.

- Development of environmentally sustainable welding strategies focusing on reduced emissions, optimized shielding gas usage, and energy-efficient process design.
- Systematic investigation of composition-dependent weld behavior across different titanium alloys, including the role of interstitial impurities and alloying elements, to establish generalized process-structure-property relationships.

• Digitalization and Process Modeling:

Artificial intelligence (AI)-based algorithms, coupled with machine learning (ML) and data-driven optimization, are expected to play a vital role in predicting weld geometry, droplet transfer behavior, and residual stress distribution. Advanced multi-physics simulations integrating arc dynamics, heat transfer, and metallurgical transformations can enable real-time control of weld quality.

• Adaptive and Intelligent Control Systems:

Closed-loop feedback control using infrared thermography, high-speed imaging, and acoustic emission can provide in-situ monitoring of droplet transfer and keyhole dynamics. Adaptive waveform control systems will allow instantaneous modulation of current, voltage, and wire feed speed to maintain thermal equilibrium during CMT and hybrid laser-arc welding.

• Additive and Multi-Material Integration:

The future of titanium welding is closely linked with Wire Arc Additive Manufacturing (WAAM). The synergy between CMT and WAAM can facilitate multi-material deposition, enabling functionally graded components with tailored microstructures and mechanical properties. This will be particularly beneficial for aerospace structures and energy systems requiring complex geometries.

• Long-term durability and fatigue qualification:

Future studies should prioritize high-cycle fatigue, low-cycle fatigue, corrosion-fatigue, creep-fatigue interaction, and damage-tolerance assessments of titanium joints produced by CMT, MIG, and Laser-MIG hybrid welding. Such studies should be performed under aerospace-relevant loading spectra and thermal environments, with direct correlation to weld toe profile, porosity level, α/β microstructural morphology, and residual stress distribution. Establishing such datasets will be essential for predicting service life and accelerating qualification

of advanced welded titanium structures in aerospace applications.

• Sustainability and Surface Engineering:

Reducing energy consumption, shielding gas usage, and emissions will be a major focus in the next decade. Development of environmentally friendly shielding systems and alloy-specific fillers will support sustainable manufacturing. Additionally, combining hybrid welding with surface modification or post-weld laser treatment can further enhance fatigue strength and corrosion resistance. **Sustainability and Environmental Impact:** Future work should emphasize quantitative evaluation of energy consumption, shielding gas usage, and emissions through lifecycle assessment (LCA) frameworks. Development of low-carbon welding processes, efficient shielding strategies, and recyclable filler materials will be critical for sustainable manufacturing.

• Standardization and Certification:

For aerospace qualification, comprehensive databases on fatigue, fracture toughness, and corrosion performance of CMT and LAMIG joints are essential. Establishing standardized welding protocols and quality assurance systems will accelerate the industrial adoption of these advanced processes.

- Integration of real-time monitoring and closed-loop feedback control for droplet transfer and keyhole dynamics.
- Development of predictive numerical models for temperature and stress distribution.
- Exploration of multi-material and graded joints using CMT-WAAM or hybrid additive welding systems.
- Detailed studies on residual stress, fatigue life, and corrosion behavior of hybrid welded joints for qualification in aerospace service.

Author Contributions

Parth Patel, Ananta Dutta, A. Ghosh, Sushovan Basak, Santosh Kumar Sahoo, and Nityananda Sahoo contributed to the conceptualization, literature review, and drafting of the manuscript. Pankaj Shrivastava and Velaphi Msomi contributed to the revision, supervision, funding acquisition and finalization of the manuscript. All authors have read and approved the final version of the manuscript.

CRediT authorship contribution statement

SHRIVASTAVA PANKAJ: Writing – review & editing, Visualization, Funding acquisition, Formal analysis. **Santosh Kumar Sahoo:** Methodology, Data curation. **Nityananda Sahoo:** Writing – review & editing, Formal analysis. **Arka Ghosh:** Writing – review & editing, Visualization, Data curation. **Sushovan Basak:** Writing – review & editing, Investigation, Data curation. **Parth Patel:** Writing – original draft, Visualization, Formal analysis, Data curation. **Ananta Dutta:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Ashutosh Das:** Visualization, Data curation. **Velaphi Msomi:** Supervision, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT (OpenAI) in order to improve the clarity of language, grammar, and overall readability of the manuscript. After using this tool, the author(s) carefully reviewed and edited the content as needed and take full responsibility for the content of the published article. No AI tools were used to generate or prepare the figures, schemes, or scientific results presented in this study.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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